Stability of coefficient minima for real stable polynomials

Astrée Iriart

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Abstract

The van der Waerden inequality for real stable polynomials identifies a unique coefficient minimizer at the balanced gradient. We prove that this minimizer persists under prescribed perturbations of the gradient, with an explicit neighborhood in every dimension. For normalized n-homogeneous real stable polynomials, the power of the corresponding linear form uniquely minimizes the all-ones coefficient whenever $\|t - \mathbf{1}\|_1 \le 10^{-9} n^{-8}$. The coefficient gap controls a root-variance deficit, with an inverse-polynomial constant after normalization by $n!/n^n$. The proof passes from a quantitative hyperbolicity-cone margin to balanced rigidity, then transports the coefficient-to-scalar ratio along logarithmic scalings. Root variance also controls the full coefficient vector, and variable replication gives weighted neighborhoods for positive integer coefficient targets. We determine the exact local stability boundary. In dimension three, a commuting-minimum theorem yields sharp global Euclidean and ℓ^1 radii, their boundary minimizers, and an escape barrier whose attainment changes at the balanced marginal.

Contribution and significance

Gurvits and Leake ask whether the balanced coefficient-minimization theorem persists when the gradient varies. We answer their Question 1.8 for the full homogeneous real stable class, and Question 1.7 for mixed discriminants, with an inverse-polynomial neighborhood. The main quantitative estimate strengthens uniqueness to variance growth. Three ingredients keep the constants explicit: a signed substitution with a nonnegative inverse, relative control of root variance under logarithmic scaling, and an exact identity for the normalized coefficient ratio. Together they turn balanced rigidity into stability under prescribed marginals.

The local and three-dimensional results describe the geometry beyond this uniform neighborhood. Centered hyperbolic roots give the exact local boundary and linear control of coefficient distance. In dimension three, a commuting reduction makes face geometry available: it determines two sharp centered global balls and all their boundary minimizers. Explicit paths then distinguish the value of an escape barrier from its attainment. The general-dimensional constants are conservative; sharpness is established for the two cubic radii.

1. Introduction

Coefficient minimization under prescribed gradients is a robustness problem for the van der Waerden inequality. At the balanced gradient, the unique minimizer among normalized homogeneous real stable polynomials is a power of the arithmetic mean. When the gradient moves, both the feasible class and the candidate minimizer move with it. We show that the minimizer persists throughout an explicit neighborhood whose radius decreases only polynomially with the degree. The resulting stability estimate also measures the distance from a polynomial to the corresponding power.

There is a second aspect to the problem. A local minimum need not remain globally minimizing as the marginal vector moves farther from balance. We determine the exact local boundary in every dimension. In dimension three, we also determine two sharp centered global balls, classify the new minimizers on their boundaries, and calculate an escape barrier. The general theorem provides uniform quantitative control; the cubic analysis shows how global competitors first appear.

1.1. The coefficient-minimization problem

A real polynomial is *real stable* if it has no zero when every variable has positive imaginary part. All stable polynomials in this paper are homogeneous and have nonnegative coefficients. For $t \in (0,\infty)^n$ with $\sum_i t_i = n$, let $\mathcal{P}_t$ consist of the degree-n real stable polynomials in n variables satisfying

$$p(\mathbf{1}) = 1, \qquad \nabla p(\mathbf{1}) = t.$$

We write

$$\ell_t(x) = \frac{t\cdot x}{n}, \qquad \mathcal{L}p = [x_1\cdots x_n]p, \qquad V_n = \frac{n!}{n^n}, \qquad d_n(t) = V_n\prod_i t_i.$$

The power ℓ_t^n belongs to $\mathcal{P}_t$ and has coefficient $d_n(t)$. Our question is when it minimizes $\mathcal{L}p$ over the entire class $\mathcal{P}_t$.

The quantity that measures departure from this power is the root-variance matrix

$$Q_p = \frac{n-1}{2n}tt^{\mathsf{T}} - \frac{1}{2}\nabla^2 p(\mathbf{1}), \qquad \delta(p) = \operatorname{tr} Q_p.$$

We prove that $Q_p \succeq 0$, $Q_p\mathbf{1} = 0$, and $\delta(p) = 0$ precisely when $p = \ell_t^n$. The terminology comes from the real roots of $s \mapsto p(s\mathbf{1}+x)$: after subtracting their mean, their squared sum is $2x^{\mathsf{T}}Q_p x$.

1.2. Global stability and its local boundary

The main global estimate is the following implication, valid for every $n \geq 3$ and every $p \in \mathcal{P}_t$:

$$\|t-\mathbf{1}\|_1 \leq \frac{1}{10^9 n^8} \quad\Longrightarrow\quad \mathcal{L}p - d_n(t) \geq \frac{V_n}{2\cdot 10^9 n^8}\,\delta(p). \tag{1.1}$$

Thus ℓ_t^n is the unique global minimizer on this neighborhood. The constants are explicit and are not asserted to be optimal. At the balanced gradient we obtain the stronger estimate

$$\frac{\mathcal{L}p}{V_n} - 1 \geq \frac{\delta(p)}{10^8 n^8}.$$

Theorem 3.1 proves both assertions. Corollary 3.8 gives a Euclidean neighborhood as well. For $n = 2$, the coefficient gap equals $\delta(p)$ for every positive gradient of sum two.

Root variance controls more than the objective coefficient. If $\|f\|_{\mathrm{coef},1}$ denotes the sum of the absolute values of the coefficients of f, then Theorem 4.1 gives an explicit K_n such that

$$\frac{4\delta(p)}{n(n-1)} \leq \|p-\ell_t^n\|_{\mathrm{coef},1} \leq K_n\delta(p).$$

Together with (1.1), this yields linear coefficient-distance stability, with the optimal distance exponent one. Equivalently, the objective gap controls total variation between the exponent law of p and the multinomial law of ℓ_t^n. The constant K_n is not claimed polynomial in n.

The exact local criterion has a simpler form. For every positive t of sum n,

$$\ell_t^n \text{ is a relative local minimum on } \mathcal{P}_t \quad \Longleftrightarrow \quad \sum_i (t_i - 1)^2 < \frac{n}{n-1}. \tag{1.2}$$

At equality a feasible perturbation has vanishing quadratic variation and a negative cubic term. The strict inequality is therefore essential. Variable replication extends both the global and local results to every positive integer target exponent α of total degree N; the corresponding local boundary is

$$\sum_i \frac{(t_i - \alpha_i)^2}{\alpha_i} = \frac{N}{N-1}.$$

The same weighted distance also appears in the explicit global neighborhood of Corollary 5.4.

1.3. The mechanism of the global proof

The proof has three stages. First, a near-minimizer must lie quantitatively inside its hyperbolicity cone. If a coordinate direction is too close to the cone boundary, a signed change of variables lowers its degree and invokes a stronger capacity inequality. The change of variables has a nonnegative inverse, so the resulting coefficient comparison is relative and preserves the natural scale V_n.

Second, this interior margin gives a probabilistic representation through a positive stochastic matrix. Conditional categorical sampling then supplies a lower bound on the Hessian of $u \mapsto \log p(e^u)$. This controls the logarithmic fields that minimize the coordinate derivative capacities. A relative perturbation estimate for Q_p allows these different fields to be compared with the original polynomial, leading to quantitative balanced rigidity.

Finally, logarithmic scaling moves a near-balanced polynomial to a balanced one. The useful quantity along this path is the ratio of its all-ones coefficient to the coefficient of the corresponding power. Its gradient is exactly $2Q_{p_u}(1/t(u) - \mathbf{1})$. The factor $1/t(u) - \mathbf{1}$ vanishes at the balanced endpoint. Integrating this identity transports the balanced gap with a small enough error to retain an inverse-polynomial neighborhood.

1.4. Mixed discriminants and sharp cubic geometry

For Hermitian positive semidefinite matrices with $\sum_i X_i = I_n$ and $\operatorname{tr} X_i = t_i$, the determinant polynomial belongs to $\mathcal{P}_t$. We use the full-coefficient normalization

$$D(X_1, \dots, X_n) = [x_1 \cdots x_n] \det\Big(\sum_i x_i X_i\Big),$$

so $D(I_n, \dots, I_n) = n!$. At the scalar tuple $X_i^0 = t_i I_n/n$, the value is $d_n(t)$, and

$$\delta(p) = \frac{1}{2} \sum_i \|X_i - X_i^0\|_F^2.$$

The global polynomial theorem therefore gives quadratic growth away from the scalar tuple. We also determine the full scalar Hessian inertia.

In dimension three, every relative local minimum consists of commuting matrices. Simultaneous diagonalization reduces the global problem to a permanent on a transportation polytope. Its faces yield the sharp centered balls

$$\sum_{i=1}^3 (t_i - 1)^2 \le \frac{3}{8}, \qquad \|t - \mathbf{1}\|_1 \le \sqrt{\frac{236 - 37\sqrt{37}}{12}}.$$

In each open ball the scalar minimizer is unique; the additional boundary minimizers and arbitrarily close outside violations are explicit. Definite ternary determinantal representations transfer these results to all normalized ternary stable cubics.

These radii expose the difference between local and global stability: the sharp squared global Euclidean radius is $3/8$, while the squared local radius is $3/2$. The scalar tuple can therefore remain a strict local minimum after it ceases to minimize globally. On a symmetric marginal family, the final section computes the least height needed to cross a prescribed quadratic level. That height varies continuously, whereas its attainment fails at the balanced marginal.

1.5. Relation to earlier results

Gurvits proved the balanced mixed-discriminant inequality and its equality classification for complex Hermitian tuples [2, Theorem 4.1]. The balanced coefficient inequality and uniqueness for real stable polynomials are [3, Corollary 2.5 and Theorem 5.7]. The degree-sensitive capacity inequality from the latter work and the capacity bound of Gurvits, Klein, and Leake [4, Theorem 2.1] are inputs to our argument. Our contribution is the quantitative near-balanced conclusion and the stability estimates described above.

Gurvits, Klein, and Leake also established unique rank-one permanent minimization under a sufficiently small perturbation of one marginal [4, Theorem 2.3]. Gurvits and Leake prove a matrix robustness theorem in which individual traces remain one and the matrix sum varies near the identity [5, Theorem 1.2]. Their Questions 1.7 and 1.8 instead fix the matrix sum, or polynomial normalization, and vary the individual traces, or gradient. Our global theorem answers these questions with an explicit inverse-polynomial radius for the full real stable class.

The sharp cubic results use more specific geometry. Helton and Vinnikov's definite ternary representation theorem [6] is used only to transfer that cubic analysis, and is not needed for the general-dimensional result. The appendix supplies the exact positivity certificate used in the ℓ^1 face calculation.

1.6. Organization

Section 2 introduces root variance, its covariance interpretation, and the local criterion. Section 3 proves the global stability estimate. Sections 4 and 5 treat coefficient distance and positive integer targets. Section 6 develops the matrix specialization. The cubic geometry occupies Sections 7–10, followed by the escape theorem in Section 11.

2. Root variance and local stability

The polynomial class admits a single measure of departure from the scalar power: the variance of its centered hyperbolic roots. We introduce this quantity first, relate it to covariance and matrix distance, and then identify the exact local stability region. The global estimate will be proved in Section 3.

For positive t with $\sum_i t_i = n$, let $\mathcal{P}_t$ consist of real stable n-homogeneous polynomials with nonnegative coefficients in n variables satisfying $p(\mathbf{1}) = 1$ and $\nabla p(\mathbf{1}) = t$. Set

$$\ell_t(x) = \frac{t \cdot x}{n}, \qquad Q_p = \frac{n-1}{2n} t t^{\mathsf{T}} - \frac{1}{2}\nabla^2 p(\mathbf{1}), \qquad \delta(p) = \operatorname{tr} Q_p.$$

We use the coefficient topology on this finite-dimensional polynomial space and write $\mathcal{L}p = [x_1 \cdots x_n]p$.

Theorem 2.1 (Stable-polynomial local and global minimization)**.** *For every $n \geq 2$ the following statements hold.*

1. *For every positive gradient vector t of sum n, the polynomial ℓ_t^n is a relative local minimum of $\mathcal{L}p$ on $\mathcal{P}_t$ if and only if $\sum_i (t_i - 1)^2 < n/(n-1)$. In this region it is a strict local minimum, with coefficient gap bounded below by a positive multiple of $\delta(p)$ in a neighborhood of ℓ_t^n.*
2. *There are $\epsilon_n, \kappa_n > 0$ such that, whenever $\|t - \mathbf{1}\|_1 < \epsilon_n$ and $p \in \mathcal{P}_t$,*

$$\mathcal{L}p - \frac{n!}{n^n} \prod_i t_i \geq \kappa_n \delta(p). \tag{2.1}$$

Here $Q_p \succeq 0$, and $\delta(p) = 0$ holds exactly when $p = \ell_t^n$. Thus the scalar power is the unique global minimizer.

Lemma 2.2 (Root variance and central coefficients)**.** *For $p \in \mathcal{P}_t$ there are homogeneous polynomials a_k of degree k such that*

$$p(x) = \ell_t(x)^n - \ell_t(x)^{n-2} q_p(x) + \sum_{k=3}^{n} \ell_t(x)^{n-k} a_k(x), \qquad q_p(x) = x^\top Q_p x. \tag{2.2}$$

The quadratic form Q_p is positive semidefinite, $Q_p \mathbf{1} = 0$, and, for every real x,

$$|a_k(x)| \leq \binom{n}{k} \big(2q_p(x)\big)^{k/2}. \tag{2.3}$$

Proof. For real x, the monic polynomial $p(s\mathbf{1}+x)$ has only real roots: a root with positive imaginary part would contradict stability, and real coefficients exclude roots in the lower half-plane as well. Write

$$p(s\mathbf{1} + x) = \prod_{i=1}^{n} (s + \lambda_i(x)), \qquad \mu_i(x) = \lambda_i(x) - \ell_t(x).$$

The coefficient of s^{n-1} is $t \cdot x$, so $\sum_i \mu_i = 0$. The elementary symmetric functions $a_k = e_k(\mu_1, \ldots, \mu_n)$ are polynomials: they are the coefficients of $p((s - \ell_t(x))\mathbf{1} + x)$. They are homogeneous of degree k, with $a_1 = 0$ and

$$a_2 = -\frac{1}{2} \sum_i \mu_i^2 = -q_p.$$

Comparing the uncentered coefficient of s^{n-2} gives exactly the displayed formula for Q_p. Hence $Q_p \succeq 0$. Euler's identity $\nabla^2 p(\mathbf{1})\mathbf{1} = (n-1)t$ gives $Q_p \mathbf{1} = 0$. Finally $|\mu_i| \leq \sqrt{2q_p(x)}$, and bounding each product in e_k proves (2.3). Setting $s = 0$ proves (2.2). No regularity of individually ordered roots is used. □

2.1. Covariance and the matrix interpretation

The same quadratic form has a probabilistic meaning. The coefficients of a normalized polynomial form a probability law on exponent vectors; the scalar power gives the corresponding multinomial law. Their covariance difference is exactly twice Q_p.

For the next proposition we allow nonnegative gradients, including zero entries. The degree and number of variables are still n, and $p(\mathbf{1}) = 1$. Put $p_t^\circ = \ell_t^n$, $f = p - p_t^\circ$, and $\delta = \operatorname{tr} Q_p$.

Proposition 2.3 (Exact Hessian and covariance identities)**.** *The centered-root matrix satisfies*

$$2Q = -\nabla^2 f(\mathbf{1}) = \frac{n-1}{n} tt^\top - \nabla^2 p(\mathbf{1}).$$

Regard the coefficients p_α as a probability distribution of an exponent vector A, and let M have the multinomial distribution with n trials and cell probabilities t/n. Then

$$\mathbb{E}A = \mathbb{E}M = t, \qquad 2Q = \operatorname{Cov}(M) - \operatorname{Cov}(A) \succeq 0.$$

In particular,

$$0 \le \delta = \frac{1}{2}\left(n - \frac{\|t\|_2^2}{n} - \operatorname{tr}\operatorname{Cov}(A)\right) \le \frac{n-1}{2}. \tag{2.4}$$

The final bound is attained by $p = x_1 \cdots x_n$. For a determinantal polynomial $p(x) = \det(\sum_i x_i X_i)$ with $X_i \succeq 0$ and $\sum_i X_i = I_n$, let $Y_i = X_i - t_i I_n/n$. Then

$$Q_{ij} = \tfrac{1}{2}\operatorname{tr}(Y_i Y_j), \qquad \delta = \tfrac{1}{2}\sum_i \|Y_i\|_F^2. \tag{2.5}$$

Proof. Set $P = I - \mathbf{1}t^\mathsf{T}/n$. The root decomposition gives $Q = -\frac{1}{2}P^\mathsf{T}\nabla^2 p(\mathbf{1})P$. Euler's identity applied to the degree-n polynomial f, whose gradient vanishes at $\mathbf{1}$, gives $\nabla^2 f(\mathbf{1})\mathbf{1} = 0$. Also $\nabla^2 p_i^\circ(\mathbf{1}) = (n-1)tt^\mathsf{T}/n$ and $t^\mathsf{T}P = 0$. Hence $P^\mathsf{T}\nabla^2 p(\mathbf{1})P = \nabla^2 f(\mathbf{1})$, proving the first identity.

Differentiation of the generating polynomial gives

$$\operatorname{Cov}(A) = \nabla^2 p(\mathbf{1}) + \operatorname{diag}(t) - tt^\mathsf{T}, \qquad \operatorname{Cov}(M) = \operatorname{diag}(t) - tt^\mathsf{T}/n.$$

Subtracting proves the covariance identity. Taking traces and using $\operatorname{tr}\operatorname{Cov}(A) \ge 0$ and $\|t\|_2^2 \ge n$ gives (2.4). For $p = x_1 \cdots x_n$, the exponent vector is constant and $t = \mathbf{1}$, giving equality.

In the determinantal case, differentiation at the identity yields $\partial_{ij} p(\mathbf{1}) = t_i t_j - \operatorname{tr}(X_i X_j)$. Substitution gives $2Q_{ij} = \operatorname{tr}(X_i X_j) - t_i t_j/n = \operatorname{tr}(Y_i Y_j)$. For Hermitian Y_i, $\operatorname{tr}(Y_i^2) = \|Y_i\|_F^2$. □

2.2. The local criterion

The central expansion isolates the quadratic contribution to the objective. Its sign determines the strict local region; the higher central coefficients are of smaller order in the variance.

Proof of Theorem 2.1. Put $C_n = (n-2)!/n^{n-2}$, $w_i = 1/t_i$, and $\delta = \operatorname{tr} Q_p$. The contribution of the quadratic term in (2.2) to the coefficient gap is

$$\begin{aligned} L_t(Q_p) &= -\mathcal{L}\big(\ell_t^{n-2} q_p\big) \\ &= C_n \prod_i t_i \left(\sum_i \frac{(Q_p)_{ii}}{t_i^2} - w^\mathsf{T} Q_p w\right). \end{aligned} \tag{2.6}$$

Indeed only the off-diagonal monomials of q_p can contribute to a squarefree coefficient. Let $R = \operatorname{diag}(1/t) Q_p \operatorname{diag}(1/t)$. Then $R \succeq 0$ and $Rt = 0$. On $t^\perp$ the smallest eigenvalue of $I - \mathbf{1}\mathbf{1}^\mathsf{T}$ is

$$\lambda(t) = \frac{n^2}{\|t\|_2^2} - (n-1).$$

This also follows from the projection calculation in Lemma 6.2. If $\lambda(t) > 0$, then

$$L_t(Q_p) \ge \alpha(t)\delta, \qquad \alpha(t) = C_n \prod_i t_i \, \frac{\lambda(t)}{(\max_i t_i)^2} > 0. \tag{2.7}$$

The higher terms admit a direct, uniform coefficient bound. For any n-homogeneous polynomial f in n variables,

$$\mathcal{L} f = \sum_{S \subseteq [n]} (-1)^{n-|S|} f(\mathbf{1}_S).$$

Inclusion–exclusion keeps precisely the monomials supported on all n variables, which in degree n are squarefree. Since $0 \le \ell_t(\mathbf{1}_S) \le 1$ and $q_p(\mathbf{1}_S) \le n\delta$, the absolute contribution of the higher terms is at most

$$2^n \sum_{k=3}^{n} \binom{n}{k} (2n\delta)^{k/2} \le B_n \delta^{3/2} \quad (0 \le \delta \le 1), \tag{2.8}$$

where $B_n = 2^n \sum_{k=3}^{n} \binom{n}{k} (2n)^{k/2}$ and $B_2 = 0$. Combining these estimates gives the explicit bound

$$\mathcal{L}(p - \ell_t^n) \ge \alpha(t)\delta - B_n \delta^{3/2} \qquad (0 \le \delta \le 1). \tag{2.9}$$

For $n \ge 3$ it is at least $\alpha(t)\delta/2$ whenever $\delta \le \min\{1, (\alpha(t)/(2B_n))^2\}$. For $n = 2$ the remainder vanishes identically. Because $Q_p \ge 0$, the equation $\delta = 0$ forces $Q_p = 0$. Lemma 2.2 then gives $p = \ell_t^n$. Also Q_p varies continuously in polynomial coefficients. This proves strict local minimality throughout the stated strict region.

For necessity, the negative commuting perturbations in the proof of Theorem 6.1 give normalized real stable polynomials

$$p_s(x) = \det\Big(\sum_i x_i X_i(s)\Big) \in \mathcal{P}_t$$

converging to ℓ_t^n with smaller all-ones coefficient, whenever $\lambda(t) < 0$ or $\lambda(t) = 0$. Their gradient is t because $\sum_i X_i(s) = I$ and $\operatorname{tr} X_i(s) = t_i$. Thus the same strict criterion is necessary in the full stable class.

For the global assertion, Theorem 3.1 shows that, for $n \ge 3$, one may take $\epsilon_n = 10^{-9} n^{-8}$ and $\kappa_n = (n!/n^n)/(2 \cdot 10^9 n^8)$. Its proof uses the root and covariance identities above, independently of the local criterion. For $n = 2$ the central expansion has no remainder and $Q_p \mathbf{1} = 0$ gives $\mathcal{L}(p - \ell_t^2) = \delta(p)$ for every positive t of sum two. This proves global uniqueness also in that case. □

The local proof also explains the role of higher central coefficients: they are negligible in the strict region, but the cubic term determines what happens at its singular boundary. The next section establishes a global neighborhood, without imposing a small-variance condition on the competing polynomial.

3. An inverse-polynomial global neighborhood

Write $\mathcal{L}p = [x_1 \cdots x_n]p$ and $V_n = n!/n^n$. The global neighborhood in Theorem 2.1 can be chosen with inverse-polynomial dependence on the dimension.

Theorem 3.1 (Polynomial global radius)**.** *Let $n \ge 3$ and $p \in \mathcal{P}_t$. If*

$$\|t - \mathbf{1}\|_1 \le \frac{1}{10^9 n^8},$$

then

$$\mathcal{L}p - V_n \prod_i t_i \ge \frac{V_n}{2 \cdot 10^9 n^8} \delta(p). \tag{3.1}$$

Consequently ℓ_t^n is the unique global minimizer on $\mathcal{P}_t$. For every balanced $p \in \mathcal{P}_{\mathbf{1}}$, one also has

$$\frac{\mathcal{L}p}{V_n} - 1 \ge \frac{\delta(p)}{10^8 n^8}. \tag{3.2}$$

The radius and constants are sufficient bounds, without an assertion of optimality. No product or determinantal representation of p is assumed.

The proof follows the three stages outlined in the introduction: a cone margin for near-minimizers, a quantitative rigidity estimate at the balanced gradient, and transport to nearby gradients. The constants remain explicit because each stage controls a relative quantity at the scale of the coefficient gap. We first record the required capacity inputs. Set

$$\operatorname{Cap}(p) = \inf_{x>0} \frac{p(x)}{x_1 \cdots x_n}, \qquad g(k) = \left(\frac{k-1}{k}\right)^{k-1} \quad (k \geq 2), \quad g(1) = 1.$$

Gurvits's degree-sensitive derivative inequality [3, Theorem 4.10] implies

$$\mathcal{L}p \geq V_n \operatorname{Cap}(p). \tag{3.3}$$

If one variable has degree at most $n-1$, eliminating that variable first improves the multiplier to $V_n g(n-1)/g(n)$. We also use the normalized capacity bound [4, Theorem 2.1]:

$$p(\mathbf{1}) = 1, \quad d = \|\nabla p(\mathbf{1}) - \mathbf{1}\|_1 < 1 \quad \Longrightarrow \quad \operatorname{Cap}(p) \geq (1-d)^n. \tag{3.4}$$

Here and below these capacity statements concern nonnegative-coefficient real stable polynomials. The stronger degree-dependent factor follows by applying the derivative inequality first to the designated variable, then to the remaining variables in the usual induction.

3.1. A polynomial margin in the hyperbolicity cone

Define

$$\eta = \frac{g(n-1)}{g(n)} - 1, \qquad r_0 = \frac{\eta}{64n^2}, \qquad m = \frac{nr_0}{8}. \tag{3.5}$$

For $n \geq 3$,

$$\frac{1}{2n^2} \leq \eta \leq 1, \qquad r_0 \geq \frac{1}{128n^4}, \qquad m \geq \frac{1}{1024n^3}. \tag{3.6}$$

Indeed, $H(x) = (x-1)\log(1-1/x)$ satisfies $H'(x) = \log(1-1/x) + 1/x \leq -1/(2x^2)$. Integration between $n-1$ and n, followed by $e^z - 1 \geq z$, gives the lower bound on η. The upper bound follows from $e^{-1} \leq g(k) \leq 1/2$ for $k \geq 2$.

We use the hyperbolicity cone containing $\mathbf{1}$. Its closure is convex, contains the nonnegative orthant, and the polynomial is hyperbolic in every interior direction; see, for example, [1, Fact 2.7]. For a vector v, let $\lambda_1(v), \ldots, \lambda_n(v)$ denote the hyperbolic eigenvalues determined by $p(z\mathbf{1} + v) = \prod_j (z + \lambda_j(v))$.

Lemma 3.2 (Interior margin)**.** *Suppose $p \in \mathcal{P}_t$, $s = \|t - \mathbf{1}\|_1 \leq \eta/(64n)$ and $\mathcal{L}p \leq V_n(1 + \eta/2)$. Then every hyperbolic eigenvalue of every e_i is greater than r_0.*

Proof. Fix i and put $r = \min_j \lambda_j(e_i) \geq 0$. Suppose $r \leq r_0$. Set $A = I - r\mathbf{1}e_i^{\mathsf{T}}$ and $q(x) = p(Ax)$. Each column of A lies in the closed cone, and the column sum is $(1-r)\mathbf{1}$, an interior point. For $y > 0$, Ay is the sum of $(\min_j y_j)(1-r)\mathbf{1}$ and a nonnegative combination of closed-cone vectors, so it is interior. Hyperbolicity in direction Ay therefore implies $p(Ax + \mathrm{i}Ay) \neq 0$ for real x. Thus q is real stable. Homogeneous real stability gives a common coefficient sign, which is positive because $q(\mathbf{1}) = (1-r)^n > 0$ [3, Fact 4.7]. Moreover $q(e_i) = 0$, so homogeneity and nonnegativity imply $\deg_{x_i} q \leq n - 1$.

The normalized gradient is

$$\nabla\left(\frac{q}{q(\mathbf{1})}\right)(\mathbf{1}) = \frac{t - nre_i}{1-r}.$$

Consequently its distance from $\mathbf{1}$ is at most

$$d \leq \frac{s + 2(n-1)r}{1-r} \leq 2s + 4nr < 1.$$

Equations (3.4) and Bernoulli's inequality give

$$\mathrm{Cap}(q) \geq (1-r)^n(1-d)^n \geq 1 - 2ns - 5n^2r.$$

The decisive comparison is provided by the nonnegative inverse

$$A^{-1} = I + \frac{r}{1-r}\mathbf{1}e_i^{\mathsf{T}}.$$

Since $p = q \circ A^{-1}$, the squarefree monomial of q contributes $\mathcal{L}q/(1-r)$ to $\mathcal{L}p$, and all other contributions are nonnegative. Using the reduced-degree version of (3.3),

$$\mathcal{L}p \geq \frac{\mathcal{L}q}{1-r} \geq \frac{V_n(1+\eta)\,\mathrm{Cap}(q)}{1-r} \geq V_n(1+\eta)(1-2ns-5n^2r).$$

But $2ns + 5n^2r \leq 7\eta/64$, and $(1+\eta)(1-7\eta/64) \geq 1 + 50\eta/64 > 1 + \eta/2$. This is the required contradiction. □

Lemma 3.3 (Uniform mixing and logarithmic conditioning)**.** *Under the hypotheses of Lemma 3.2, write $F(u) = \log p(e^{u_1}, \ldots, e^{u_n})$. For $u \perp \mathbf{1}$ with $\|u\|_2 \leq 1$,*

$$\nabla^2 F(u) \geq mI \quad \text{on } \mathbf{1}^{\perp}. \tag{3.7}$$

For arbitrary $u \perp \mathbf{1}$, if $t = \mathbf{1}$ and $F(u) \leq \tau \leq r_0/(64n)$, then

$$\|u\|_2 \leq 4\sqrt{\frac{\tau}{nr_0}}. \tag{3.8}$$

Proof. Set $a = 1 - nr_0$, $B = I - r_0\mathbf{1}\mathbf{1}^{\mathsf{T}}$ and $M = aI + r_0\mathbf{1}\mathbf{1}^{\mathsf{T}}$. The preceding cone argument applied simultaneously to all columns shows that $h(x) = p(Bx)/a^n$ is normalized, real stable and nonnegative. Since $BM = aI$, we have $p(x) = h(Mx)$. Every row of M is stochastic with each entry at least r_0.

The coefficient law of p is obtained by sampling the exponent vector of h and independently routing its n units according to the corresponding rows of M. After exponential tilting by u, the conditional routing probabilities are

$$\pi_{ij}(u) = \frac{M_{ij}e^{u_j}}{\sum_k M_{ik}e^{u_k}} \geq r_0e^{-2} \geq r_0/8 \quad (\|u\|_2 \leq 1).$$

For $v \perp \mathbf{1}$, one categorical draw has variance at least $(r_0/8)\min_c \sum_j (v_j - c)^2 = (r_0/8)\|v\|_2^2$. Conditional covariance of the n independent draws, followed by the law of total covariance, proves (3.7).

When $t = \mathbf{1}$, the tangent gradient of F at zero vanishes. Hence $F(u) \geq m\|u\|_2^2/2$ in the unit ball. Convexity along rays gives $F(u) \geq m\|u\|_2/2$ outside that ball. The assumed bound on τ first forces $\|u\|_2 < 1$, then yields (3.8). □

3.2. Relative root-variance control

For logarithmic tilting write

$$p_u(x) = \frac{p(e^{u_1}x_1, \ldots, e^{u_n}x_n)}{p(e^u)}, \qquad t(u) = \nabla p_u(\mathbf{1}).$$

The coefficient law is an exponential family. Its covariance matrix is

$$C(u) = \mathrm{diag}(t(u)) - \frac{t(u)t(u)^{\mathsf{T}}}{n} - 2Q_{p_u}. \tag{3.9}$$

In particular $0 \preceq C(u) \preceq \mathrm{diag}(t(u))$ and $0 \leq \delta(p_u) \leq (n-1)/2$.

Lemma 3.4 (Relative tilt distortion)**.** *For every normalized nonnegative homogeneous real stable polynomial of degree $n \geq 3$ in n variables and every real u, along the path p_{vu} one has*

$$\left|\frac{d}{dv}\delta(p_{vu})\right| \leq 12n\|u\|_\infty \delta(p_{vu}), \tag{3.10}$$

$$\left|\frac{d}{dv}(Q_{p_{vu}})_{ii}\right| \leq 12n\|u\|_\infty \delta(p_{vu}). \tag{3.11}$$

Consequently the endpoint deficits differ by a multiplicative factor at most $\exp(12n\|u\|_\infty)$.

Proof. All quantities in the following calculation are evaluated at the current point of the path. Put $b = t \cdot u$, $U = \|u\|_\infty$ and $Q = Q_{p_{vu}}$. The centered cubic a_3 from Lemma 2.2 satisfies

$$a_3(z) = \frac{1}{3}\sum_j \mu_j(z)^3, \qquad |a_3(z)| \leq \frac{1}{3}(2z^{\mathsf{T}}Qz)^{3/2}.$$

Cubic real polarization, applied to the seminorm $\|z\|_Q = (z^{\mathsf{T}}Qz)^{1/2}$, gives for $T_3 = D^3 a_3$

$$|T_3[v,w,z]| \leq 18\sqrt{2}\,\|v\|_Q\|w\|_Q\|z\|_Q.$$

The estimate follows first for $Q+\varepsilon I$ from the polarization constant $3^3/3!$, then by letting ε decrease to zero. Differentiation of the normalized coefficients gives

$$\begin{aligned} t_i' &= t_i(u_i - b/n) - 2(Qu)_i, \\ Q_{ii}' &= 2(u_i - b/n)Q_{ii} - \frac{2}{n}t_i(Qu)_i - \frac{1}{2}T_3[e_i,e_i,u]. \end{aligned}$$

For clarity, the third derivative entering the latter equality is

$$\begin{aligned} D^3 p_{vu}(\mathbf{1})[e_i,e_i,u] = {} & \frac{(n-1)(n-2)}{n^2}t_i^2 b \\ & - \frac{2(n-2)}{n}\{2t_i(Qu)_i + bQ_{ii}\} + T_3[e_i,e_i,u]. \end{aligned}$$

The three terms in Q_{ii}' are bounded respectively by $4U\delta$, $2\sqrt{n}\,U\delta$ and $9nU\delta$. Here $u^{\mathsf{T}}Qu \leq nU^2\delta$, $t_i \leq n$ and $\delta \leq (n-1)/2$. Thus (3.11) follows from $4 + 2\sqrt{n} + 9n \leq 12n$.

For the trace, sum the exact formula before estimating. The middle term becomes $-(2/n)t^{\mathsf{T}}Qu$, with the same bound $2\sqrt{n}\,U\delta$ because $\|t\|_2 \leq n$. The sum of the cubic terms is bounded by $9nU\delta$. This proves (3.10). Gronwall's inequality proves the endpoint statement, including $\delta = 0$ without division by δ. □

Lemma 3.5 (Refined one-variable capacity loss)**.** *Let $f(z) = f(0)\prod_{j=1}^n (1+b_j z)$ with $b_j > 0$, and let $z_* > 0$ minimize $f(z)/z$. Define $a_j = b_j z_*/(1+b_j z_*)$. Then $\sum_j a_j = 1$ and*

$$\frac{f'(0)}{\inf_{z>0} f(z)/z} \geq g(n)\left(1 + \frac{\sum_j a_j^2 - 1/n}{128}\right). \tag{3.12}$$

Proof. Put $S_0 = \sum_j b_j$, $v_j = b_j/S_0$, $w = \sum_j (v_j - 1/n)^2$ and $z_0 = n/((n-1)S_0)$. Direct substitution gives

$$\frac{f(z_0)}{z_0} = \frac{f'(0)}{g(n)}\prod_j (1 + v_j - 1/n) \leq \frac{f'(0)}{g(n)}e^{-w/8},$$

where $\log(1+x) \leq x - x^2/8$ for $-1/2 \leq x \leq 1$ was used. Consequently the left side of (3.12) is at least $g(n)(1 + w/8)$.

Write $S = S_0 z_*$. Stationarity gives $1 = \sum_j Sv_j/(1+Sv_j) \geq S/(1+S\max_j v_j)$. If $w \leq 1/16$, then $\max v_j \leq 1/n + 1/4 \leq 3/4$, so $S \leq 4$. The map $x \mapsto Sx/(1+Sx)$ is therefore 4-Lipschitz. Projection onto constants yields

$$\sum_j (a_j - 1/n)^2 \leq \sum_j \left(a_j - \frac{S}{n+S}\right)^2 \leq 16w.$$

For $w \geq 1/16$ the same conclusion follows from $\sum_j (a_j - 1/n)^2 \leq 1 \leq 16w$. This proves (3.12). □

Proposition 3.6 (Normalized balanced rigidity)**.** *Every $p \in \mathcal{P}_1$ satisfies* (3.2).

Proof. Set $\tau = \mathcal{L}p/V_n - 1 \geq 0$. Balanced capacity is one: Jensen's inequality for the coefficient law gives $p(e^u) \geq e^{\sum_i u_i}$. Let $T = r_0/(160000n^3)$. Suppose first that $\tau \leq T$. Lemma 3.2 applies, and in particular $p(e_j) > r_0^n$ for every j. For each i, let $q_i = \partial_i p|_{x_i=0}$. Its pure coefficients in the other variables are positive: the bivariate restriction to x_i, x_j has positive endpoint coefficients, and Newton's normalized coefficient inequalities give $[x_i x_j^{n-1}]p \geq nr_0^n$. Thus the capacity of q_i is attained at some positive y with $\prod_{j\neq i} y_j = 1$; in logarithmic coordinates its positive pure powers make sublevel sets compact.

The degree-$n-1$ capacity inequality gives

$$\mathrm{Cap}(q_i) \leq \frac{\mathcal{L}p}{V_{n-1}} = g(n)(1+\tau).$$

For $f(z) = p(y,z)$, choose z_* minimizing $f(z)/z$. Its constant and leading coefficients are positive. The usual one-variable capacity inequality yields

$$1 \leq \inf_{z>0} \frac{f(z)}{z} \leq \frac{\mathrm{Cap}(q_i)}{g(n)} \leq 1+\tau.$$

Let u_i be the centered logarithm of the vector (y, z_*). By homogeneity, $p(e^{u_i}) = \inf f(z)/z$. Lemma 3.3 gives

$$R_i = \|u_i\|_2 \leq 4\sqrt{\frac{\tau}{nr_0}} \leq \frac{1}{100n^2}.$$

The ith gradient coordinate of p_{u_i} is one by stationarity. Lemma 3.5, together with the preceding capacity bounds, gives

$$(Q_{p_{u_i}})_{ii} = \frac{1}{2}\left(\sum_j a_j^2 - \frac{1}{n}\right) \leq 64\tau.$$

The optimizing fields may depend on i. Since $\exp(12nR_i) < 2$, Lemma 3.4 nevertheless gives

$$|(Q_p)_{ii} - (Q_{p_{u_i}})_{ii}| \leq 24nR_i\delta(p).$$

After summing, $\delta(p) \leq 64n\tau + 0.24\delta(p)$, hence $\delta(p) \leq 128n\tau$.

If $\tau > T$, use $\delta(p) \leq (n-1)/2$ instead. In both cases

$$\tau \geq \min\left\{\frac{1}{128n}, \frac{2T}{n-1}\right\}\delta(p) \geq \frac{\delta(p)}{10^8 n^8},$$

where (3.6) proves the final comparison. □

The absorption argument is the point at which the coordinatewise capacity estimates become a single global statement. Each optimizing field may be different, but its perturbation error is proportional to the original total variance, so the errors can be absorbed after summation.

3.3. Transport of the normalized coefficient ratio

Lemma 3.7 (Ratio-gradient identity)**.** *If $\mathcal{L}p > 0$ and $t(u) > 0$, then*

$$\nabla_u \log \frac{\mathcal{L}p_u}{V_n \prod_i t_i(u)} = 2Q_{p_u}\big(1/t(u) - \mathbf{1}\big), \tag{3.13}$$

where reciprocals are componentwise.

Proof. The gradient of $\log \mathcal{L}p_u$ is $\mathbf{1} - t(u)$, and the Jacobian of $t(u)$ is $C(u)$. Thus the left side is $\mathbf{1} - t - C(1/t)$. Equation (3.9) gives $C(1/t) = \mathbf{1} - t - 2Q(1/t)$, and $Q\mathbf{1} = 0$ finishes the calculation. □

Proof of Theorem 3.1. The balanced assertion is Proposition 3.6. Put $s = \|t - \mathbf{1}\|_1$ and $\beta = 1/(10^8 n^8)$. If $\mathcal{L}p > V_n(1 + \eta/2)$, then $\prod_i t_i \le 1$ and $\delta < n/2$, so

$$\mathcal{L}p - V_n \prod_i t_i > V_n \eta/2 \ge V_n \eta\delta/n,$$

which is stronger than (3.1). In the other case Lemmas 3.2 and 3.3 apply.

We first work under the sufficient conditions

$$s \le \min\left\{\frac{\eta}{64n}, \frac{m}{24n}, m\sqrt{\frac{\beta}{64n}}\right\}. \tag{3.14}$$

We first construct a short logarithmic path to the balanced class. On $\mathbf{1}^\perp$, the function $F(u) = \log p(e^u)$ has tangent gradient $t - \mathbf{1}$ at zero, of norm at most s, and Hessian at least mI in the unit ball. Its boundary has $F(u) \ge -s + m/2 > F(0)$. A minimizer on the ball is therefore interior, hence is a global minimizer by convexity. Call it u_*. The polynomial p_{u_*} is balanced, and strong monotonicity gives

$$R = \|u_*\|_2 \le s/m \le 1/(24n), \qquad R^2 \le \beta/(64n).$$

The capacity bound (3.4) ensures $\mathcal{L}p > 0$. Thus all logarithms below are defined.

We estimate the ratio along this path from its balanced endpoint. Along vu_*, covariance is bounded above by nI by (3.9). Integrating from the balanced endpoint gives

$$\|t(vu_*) - \mathbf{1}\|_2 \le n(1 - v)R, \quad t_i(vu_*) \ge 23/24, \quad \|1/t(vu_*) - \mathbf{1}\|_2 \le 2n(1 - v)R.$$

Write $\delta_* = \delta(p_{u_*})$ and $H(u) = \log(\mathcal{L}p_u/(V_n \prod_i t_i(u)))$. Lemma 3.4 gives $\delta(p_{vu_*}) \le 2\delta_*$ and $\delta_* \ge \delta(p)/2$. Using (3.13) and $\|Q\|_{\mathrm{op}} \le \operatorname{tr} Q$,

$$|H(0) - H(u_*)| \le 16nR^2\delta_* \le \beta\delta_*/4.$$

By balanced rigidity and $\beta\delta_* \le 1/2$,

$$H(u_*) \ge \log(1 + \beta\delta_*) \ge \beta\delta_*/2.$$

The balanced lower bound therefore survives transport: $H(0) \ge \beta\delta(p)/8$. Finally, $\prod_i t_i \ge \prod_i (1 - |t_i - 1|) \ge 1 - s \ge 1/2$, whence

$$\mathcal{L}p - V_n \prod_i t_i = V_n \prod_i t_i \,(e^{H(0)} - 1) \ge V_n \beta\delta(p)/16 \ge \frac{V_n \delta(p)}{2 \cdot 10^9 n^8}.$$

By (3.6), the three terms in (3.14) are at least

$$\frac{1}{128n^3}, \qquad \frac{1}{24576n^4}, \qquad \frac{1}{81920000n^{15/2}},$$

respectively. Each exceeds $1/(10^9 n^8)$ for $n \ge 3$. This proves the stated radius. Rigidity at $\delta = 0$ proves uniqueness. □

Corollary 3.8 (A Euclidean neighborhood)**.** *For $n \geq 3$, inequality (3.1) and uniqueness also hold whenever $p \in \mathcal{P}_t$ and*

$$\|t - \mathbf{1}\|_2^2 \leq \frac{1}{10^{16} n^{15}}.$$

Proof. Put $r = \|t - \mathbf{1}\|_2$. The proof of Theorem 3.1 uses the ℓ^1 norm only for spectral peeling and the final bound on $\prod_i t_i$; both follow from $s \leq \sqrt{n}\, r$. For the interior balancing field, strong monotonicity gives the sharper bound $R \leq r/m$ directly. It therefore suffices that

$$r \leq \min\left\{ \frac{\eta}{64n\sqrt{n}}, \frac{m}{24n}, m\sqrt{\frac{\beta}{64n}} \right\}.$$

By (3.6), the three right-hand terms are at least $1/(128n^{7/2})$, $1/(24576n^4)$ and $1/(81920000n^{15/2})$, respectively. Each is at least $1/(10^8 n^{15/2})$. The remaining ratio-transport argument is unchanged. □

4. From root variance to coefficient distance

Root variance is determined by the Hessian at $\mathbf{1}$. Real stability makes this second-order quantity strong enough to control the entire coefficient vector. We prove a global comparison first, then combine it with the preceding coefficient-gap theorem.

For a real homogeneous polynomial $g(x) = \sum_{|\alpha|=n} g_\alpha x^\alpha$, write $\|g\|_{\mathrm{coef},1} = \sum_\alpha |g_\alpha|$. Let p be homogeneous real stable of degree $n \geq 2$ in n variables, with nonnegative coefficients, $p(\mathbf{1}) = 1$, and $\nabla p(\mathbf{1}) = t$. Thus $t_i \geq 0$ and $\sum_i t_i = n$. Put

$$\ell_t(x) = \frac{t \cdot x}{n}, \qquad p_t^\circ = \ell_t^n, \qquad f = p - p_t^\circ.$$

The centered-root decomposition defines a positive semidefinite matrix Q with $Q\mathbf{1} = 0$. We use $\delta = \operatorname{tr} Q$.

Theorem 4.1 (Global comparison of coefficient distance and deficit)**.** *For every polynomial in the preceding class,*

$$\frac{4\delta}{n(n-1)} \leq \|p - p_t^\circ\|_{\mathrm{coef},1} \leq K_n \delta, \tag{4.1}$$

where the explicit, nonoptimal constant

$$K_n = n + 2\sum_{k=3}^{n} \binom{n}{k} \frac{(kn)^k}{k!} (n-1)^{\lceil (k-2)/2 \rceil}$$

depends only on n. An empty sum is zero, so $K_2 = 2$. No lower bound on the positive entries of t is needed.

Proof. Write $f = \sum_\alpha f_\alpha x^\alpha$ and $w_\alpha = \sum_i \alpha_i(\alpha_i - 1)$. The Hessian identity gives $2\delta = -\sum_\alpha w_\alpha f_\alpha$. Since $\sum_\alpha f_\alpha = f(\mathbf{1}) = 0$ and $0 \leq w_\alpha \leq n(n-1)$, subtracting the midpoint of this interval from every w_α gives

$$2\delta \leq \frac{n(n-1)}{2} \sum_\alpha |f_\alpha|.$$

This proves the first inequality.

For the upper bound, set $P = I - \mathbf{1}t^{\mathsf{T}}/n$. The centered-root expansion is

$$f(x) = -\ell_t(x)^{n-2} x^{\mathsf{T}} Q x + \sum_{k=3}^{n} \ell_t(x)^{n-k} h_k(x).$$

Here h_k are the central coefficients of Lemma 2.2. The roots of $s \mapsto p(s\mathbf{1} + Px)$ are real and centered, and their squared sum is $2x^{\mathsf{T}}Qx$. The elementary-symmetric coefficient formula therefore gives

$$|h_k(x)| \le \binom{n}{k}(2x^{\mathsf{T}}Qx)^{k/2} \le \binom{n}{k}(2\delta)^{k/2}\|x\|_2^k \qquad (x \in \mathbb{R}^n). \tag{4.2}$$

We make the passage to coefficients explicit. If h is homogeneous of degree k and $|h(x)| \le B\|x\|_2^k$ for real x, let T be its symmetric polarization. The real polarization identity is

$$T(v_1, \dots, v_k) = \frac{1}{2^k k!} \sum_{\varepsilon \in \{-1,1\}^k} \Big(\prod_j \varepsilon_j\Big) h\Big(\sum_j \varepsilon_j v_j\Big).$$

For coordinate unit vectors this has absolute value at most $Bk^k/k!$. Multiplying by the corresponding multinomial coefficients and summing them, whose total is n^k, yields

$$\|h\|_{\mathrm{coef},1} \le B\frac{(kn)^k}{k!}.$$

Moreover, positivity of Q gives

$$\|x^{\mathsf{T}}Qx\|_{\mathrm{coef},1} = \sum_{i,j}|Q_{ij}| \le \left(\sum_i \sqrt{Q_{ii}}\right)^2 \le n\delta.$$

The coefficient norm is submultiplicative, and $\|\ell_t^j\|_{\mathrm{coef},1} = 1$ because its coefficients are nonnegative and $\ell_t(\mathbf{1}) = 1$. Thus

$$\|f\|_{\mathrm{coef},1} \le n\delta + \sum_{k=3}^{n}\binom{n}{k}\frac{(kn)^k}{k!}(2\delta)^{k/2}.$$

By (2.4),

$$(2\delta)^{k/2} \le 2\delta(n-1)^{(k-2)/2} \le 2\delta(n-1)^{\lceil (k-2)/2\rceil}.$$

This is the asserted bound, including $\delta = 0$ by continuity (or directly by rigidity of the centered roots). □

Corollary 4.2 (Linear coefficient and total-variation stability)**.** *For every fixed $n \ge 2$, there exist $\epsilon_n, c_n > 0$ such that if $\|t - \mathbf{1}\|_1 < \epsilon_n$, then every normalized homogeneous real-stable polynomial of degree n with gradient t satisfies*

$$[x_1 \cdots x_n]p - \frac{n!}{n^n}\prod_i t_i \ge c_n\|p - p_t^\circ\|_{\mathrm{coef},1}.$$

Equivalently, the coefficient gap is at least $2c_n$ times the total variation distance between the exponent law of p and the multinomial law with parameters $(n, t/n)$. The exponent one on the coefficient distance cannot be decreased.

Proof. Theorem 2.1 gives a uniform gap of at least $\kappa_n\delta$. Apply Theorem 4.1 with $c_n = \kappa_n/K_n$. Theorem 3.1 makes both ε_n and κ_n explicit; K_n itself is not claimed polynomial in n.

The two probability laws have coefficient vectors p and p_t°, so their total variation distance is half the displayed coefficient norm. To verify the optimal exponent, fix a positive gradient vector in the local stability region and a nonzero real v with $v \cdot \mathbf{1} = 0$. For all sufficiently small real s, the linear forms $\ell_t \pm s(v \cdot x)$ have nonnegative coefficients. Hence

$$p_s = (\ell_t + s(v\cdot x))(\ell_t - s(v\cdot x))\ell_t^{n-2} = p_t^\circ - s^2\ell_t^{n-2}(v\cdot x)^2$$

is normalized, real stable, and has gradient t. Its coefficient distance is exactly $s^2\|\ell_t^{n-2}(v\cdot x)^2\|_{\mathrm{coef},1}$, whereas its coefficient gap is exactly a positive constant times s^2 by positivity of the leading root-deficit functional. Consequently no positive uniform bound by the θth power of this distance can hold for $\theta < 1$. □

5. Arbitrary positive multi-index coefficients

A coefficient with a positive integer exponent vector can be made squarefree by replacing each variable with an average of new variables. This substitution preserves stability and carries the gradient distance exactly. It gives the global theorem and its quantitative bounds; the local converse requires perturbations that respect the replication blocks. In this section $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{Z}^m_{>0}$, $N = \sum_i \alpha_i \geq 2$, and $t \in \mathbb{R}^m_{>0}$ satisfies $\sum_i t_i = N$. Write $\alpha! = \prod_i \alpha_i!$ and

$$b_\alpha(t) = \frac{N!}{\alpha!}\prod_{i=1}^m (t_i/N)^{\alpha_i}, \qquad k_\alpha = \prod_{i=1}^m \frac{\alpha_i!}{\alpha_i^{\alpha_i}}.$$

Let p range over homogeneous real stable degree-N polynomials in m variables with nonnegative coefficients, $p(\mathbf{1}) = 1$, and $\nabla p(\mathbf{1}) = t$. Its scalar polynomial is $p_t(x) = \ell_t(x)^N$, where $\ell_t(x) = t \cdot x/N$. Define the root-variance matrix and its weighted trace by

$$Q_p = \frac{N-1}{2N} tt^\top - \frac{1}{2}\nabla^2 p(\mathbf{1}), \qquad \delta_\alpha(p) = \sum_i \frac{(Q_p)_{ii}}{\alpha_i}.$$

The root-variance lemma applies in m variables and degree N: $Q_p \succeq 0$, $Q_p\mathbf{1} = 0$, and $Q_p = 0$ if and only if $p = p_t$.

Theorem 5.1 (Multi-index minimization and exact local stability)**.** *Let $\varepsilon_N, \kappa_N > 0$ be constants in the squarefree near-balanced variance-growth theorem in degree N. If $\|t - \alpha\|_1 < \varepsilon_N$, then every p as above satisfies*

$$[x^\alpha]p - b_\alpha(t) \;\geq\; \frac{\kappa_N}{k_\alpha}\,\delta_\alpha(p). \tag{5.1}$$

In particular, p_t is the unique coefficient minimizer. For every fixed positive t, the scalar polynomial is a relative local minimum in coefficient topology if and only if

$$\sum_i \frac{(t_i - \alpha_i)^2}{\alpha_i} < \frac{N}{N-1}. \tag{5.2}$$

In the strict region it has local positive growth proportional to δ_α. At equality, whenever such a positive t exists, a product of positive linear forms approaching p_t has a strictly smaller target coefficient. The case $m = 1$ is a singleton and the strict inequality holds automatically.

Proof. Introduce α_i variables y_{ij} for each i, and substitute

$$x_i = \frac{1}{\alpha_i}\sum_{j=1}^{\alpha_i} y_{ij} \quad \text{in } p(x).$$

Call the resulting degree-N, N-variable polynomial P. An average of numbers in the upper half-plane stays in that half-plane, so P is real stable. Its normalization is unchanged, and its gradient has entries $\widetilde{t}_{ij} = t_i/\alpha_i$. In particular,

$$\|\widetilde{t} - \mathbf{1}\|_1 = \|t - \alpha\|_1, \qquad \sum_{i,j}\widetilde{t}_{ij}^2 = \sum_i t_i^2/\alpha_i.$$

Only x^α can contribute to the coefficient containing all N cloned variables. Thus

$$[\prod_{i,j} y_{ij}]P = k_\alpha [x^\alpha]p. \tag{5.3}$$

Let B be the averaging matrix of this substitution. The root-variance matrices transform by $Q_P = B^\top Q_p B$, whence

$$\operatorname{tr} Q_P = \sum_i (Q_p)_{ii}/\alpha_i = \delta_\alpha(p). \tag{5.4}$$

The squarefree theorem, divided by k_α, proves (5.1). If equality with the scalar value holds, $Q_P = 0$, so P is the scalar power. Setting every variable in each block equal to x_i recovers $p = p_t$.

We next prove the local statement. The exact local theorem for N variables gives sufficiency of $\sum_i t_i^2/\alpha_i < N^2/(N-1)$, which is equivalent to (5.2). It also gives local variance growth by (5.4). Necessity requires care: a general perturbation in the cloned space need not descend. We now give perturbations that are constant within each block.

Put $T = \sum_i t_i^2/\alpha_i$. When $T > N^2/(N-1)$, set

$$z_i = 1 - \frac{N t_i}{\alpha_i T}, \qquad d_i = t_i z_i.$$

Then $\sum_i d_i = 0$. Choose any real $h_1, \dots, h_N$ with $\sum_k h_k = 0$ and $\sum_k h_k^2 > 0$, and consider

$$p_s(x) = \prod_{k=1}^{N} \bigl(\ell_t(x) + s h_k\, d \cdot x\bigr). \tag{5.5}$$

For sufficiently small $|s|$, every linear form has strictly positive coefficients. Hence p_s is stable, normalized, and has gradient t. Let z^* repeat each z_i exactly α_i times and set $L = N - N^2/T > 1$. Then $\sum_r z_r^* = \sum_r (z_r^*)^2 = L$, so $e_2(z^*) = L(L-1)/2 > 0$. The quadratic coefficient in $[x^\alpha]p_s$ is

$$e_2(h)\, b_\alpha(t)\, \frac{N^2}{\binom{N}{2}} e_2(z^*) < 0,$$

since $e_2(h) = -\frac{1}{2}\sum_k h_k^2 < 0$. This disproves local minimality directly in the original variable space.

At the boundary the quadratic variation vanishes, so a cubic direction is needed. For $T = N^2/(N-1)$, put instead $z_i = 1 - (N-1)t_i/(N\alpha_i)$ and again $d_i = t_i z_i$. Let z^* denote the length-N vector in which z_i occurs α_i times. Then

$$\sum_{r=1}^{N} z_r^* = \sum_{r=1}^{N} (z_r^*)^2 = 1, \qquad z_r^* < 1.$$

It follows that $e_2(z^*) = 0$ and

$$e_3(z^*) = \frac{\sum_r (z_r^*)^3 - 1}{3} < 0.$$

This equality case is impossible when $N = 2$ and $t > 0$; for $N \geq 3$ choose $h = (N-1, -1, \dots, -1)$ in (5.5). The linear and quadratic coefficient variations vanish. Direct expansion, or (5.3), gives the cubic coefficient

$$\frac{2(N-3)!}{N^{N-3}} \frac{\prod_i t_i^{\alpha_i}}{\alpha!}\, e_3(z^*) \sum_k h_k^3 < 0,$$

since $\sum_k h_k^3 = N(N-1)(N-2) > 0$. Small positive s therefore strictly lowers the coefficient. All these perturbations are polynomials in the original m variables, completing necessity. □

Remark 5.2 (Covariance interpretation). If $p(x) = \sum_\beta p_\beta x^\beta$, regard (p_β) as the probability mass function of a random exponent vector Z. Then $\mathbb{E}Z = t$ and

$$2Q_p = \operatorname{diag}(t) - tt^\top/N - \operatorname{Cov}(Z).$$

Thus the right side of (5.1) is a positive multiple of

$$\delta_\alpha(p) = \frac{1}{2}\sum_i \frac{t_i - t_i^2/N - \operatorname{Var}(Z_i)}{\alpha_i}.$$

This is the weighted trace of the covariance deficit from the multinomial exponent distribution of ℓ_t^N. The constants are uniform over all positive compositions α of the same fixed degree N, apart from the displayed factor k_α^{-1}; no dimension-independent neighborhood is asserted.

Corollary 5.3 (Multi-index coefficient-distance stability)**.** *Let $\varepsilon_N, c_N > 0$ be constants in the squarefree linear coefficient-distance theorem. If $\|t - \alpha\|_1 < \varepsilon_N$, then*

$$[x^\alpha]p - b_\alpha(t) \geq \frac{c_N}{k_\alpha}\|p - \ell_t^N\|_{\mathrm{coef},1}.$$

Equivalently, this is a lower bound by $2c_N/k_\alpha$ times the total variation distance between the exponent distribution of p and the multinomial distribution with parameters $(N, t/N)$.

Proof. The block-average substitution is an isometry for the coefficient ℓ^1 norm, even on signed polynomials. Indeed, each monomial x^β maps to a polynomial with nonnegative coefficients summing to one; its cloned monomials have block degree β_i in block i. Distinct original multi-indices therefore give disjoint cloned supports, so no cancellation between them is possible. Applying the squarefree coefficient-distance theorem to the cloned polynomial and using (5.3) proves the assertion. The total variation statement follows by dividing the coefficient norm by two. □

The Euclidean refinement takes a particularly natural form after replication: its squared distance is the same weighted quadratic expression that describes the local boundary.

Corollary 5.4 (Explicit weighted Euclidean neighborhood)**.** *In the notation of this section, let $N \geq 3$. If*

$$\sum_i \frac{(t_i - \alpha_i)^2}{\alpha_i} \leq \frac{1}{10^{16}N^{15}},$$

then

$$[x^\alpha]p - b_\alpha(t) \geq \frac{N!/N^N}{2 \cdot 10^9 N^8 k_\alpha}\,\delta_\alpha(p).$$

In particular the scalar power is the unique global minimizer. The same growth conclusion holds under $\|t - \alpha\|_1 \leq 10^{-9}N^{-8}$.

Proof. The cloned gradient satisfies $\|\widetilde{t} - \mathbf{1}\|_2^2 = \sum_i (t_i - \alpha_i)^2/\alpha_i$. Apply Corollary 3.8, then divide by k_α using (5.3) and (5.4). The ℓ^1 assertion follows from Theorem 3.1 and the exact gradient identity in the cloning proof. □

6. Mixed discriminants and the scalar Hessian

For determinant polynomials, root variance is a matrix distance. The global result is therefore already a consequence of the polynomial theorem. The remaining local calculation gives the full Hessian inertia and explicit commuting descent directions, including the degenerate boundary.

Let $n \geq 2$, let $t = (t_1, \dots, t_n) \in (0, \infty)^n$ satisfy $\sum_i t_i = n$, and write

$$\mathcal{F}_t = \{(X_1, \dots, X_n) : X_i = X_i^* \succeq 0,\ \operatorname{tr} X_i = t_i,\ \textstyle\sum_i X_i = I_n\}.$$

All matrix spaces in this section are regarded as real vector spaces. We use the coefficient normalization

$$D(X_1, \dots, X_n) = [z_1 \cdots z_n] \det\Big(\sum_{i=1}^n z_i X_i\Big),$$

so that $D(I_n, \dots, I_n) = n!$. Put

$$X^0(t) = \Big(\frac{t_1}{n} I_n, \dots, \frac{t_n}{n} I_n\Big), \qquad d_n(t) = D(X^0(t)) = \frac{n!}{n^n} \prod_i t_i.$$

For a tuple A we set $\|A\|^2 = \sum_i \operatorname{tr}(A_i^2)$. The tangent space at every scalar tuple is the same space

$$\mathcal{E} = \{(A_i)_{i=1}^n : A_i = A_i^*, \operatorname{tr} A_i = 0,\ \textstyle\sum_i A_i = 0\}.$$

Positive traces ensure that $X^0(t)$ is in the relative interior of $\mathcal{F}_t$. In particular, every $A \in \mathcal{E}$ gives a feasible line $X^0(t) + sA$ for all sufficiently small real s.

Theorem 6.1 (Exact local stability)**.** *For every $n \geq 2$ and every positive trace vector t of sum n, the scalar tuple $X^0(t)$ is a relative local minimum of D on $\mathcal{F}_t$ if and only if*

$$\sum_{i=1}^n (t_i - 1)^2 < \frac{n}{n-1}. \tag{6.1}$$

When (6.1) holds, there are $a(t) > 0$ and $r(t) > 0$ such that

$$D(X^0(t) + A) - d_n(t) \geq a(t)\|A\|^2 \quad (A \in \mathcal{E},\ \|A\| < r(t)).$$

Above the threshold the scalar tuple has a negative second variation. At equality, whenever such a positive trace vector exists, a commuting feasible perturbation has vanishing linear and quadratic terms and a strictly negative cubic term. Equality and the region above it are infeasible for $n = 2$.

Lemma 6.2 (Hessian and inertia)**.** *Let $T = \sum_i t_i^2$ and $\lambda(t) = n^2/T - (n-1)$. The scalar tuple is stationary on its affine constraint space, and its Hessian is*

$$\mathcal{H}_t(A) = \frac{(n-2)!}{n^{n-2}} \prod_i t_i \left(\sum_i \left\| \frac{A_i}{t_i} \right\|_F^2 - \left\| \sum_i \frac{A_i}{t_i} \right\|_F^2 \right). \tag{6.2}$$

Writing $h = n^2 - 1$, its inertia, in the order (positive, negative, zero), is

$$\begin{cases} ((n-1)h, 0, 0), & \lambda(t) > 0, \\ ((n-2)h, 0, h), & \lambda(t) = 0, \\ ((n-2)h, h, 0), & \lambda(t) < 0. \end{cases}$$

For real symmetric tuples the same statement holds with $h = n(n+1)/2 - 1$.

Proof. Polarizing the first two coefficients of $\det(I + sB)$ gives

$$D(B, I, \dots, I) = (n-1)! \operatorname{tr} B,$$
$$D(B, C, I, \dots, I) = (n-2)!\big(\operatorname{tr} B \operatorname{tr} C - \operatorname{tr}(BC)\big).$$

The first variation vanishes since every A_i is traceless. By multilinearity, the second variation is

$$-2\frac{(n-2)!}{n^{n-2}}\prod_i t_i \sum_{i<j}\frac{\operatorname{tr}(A_iA_j)}{t_it_j},$$

which is (6.2).

Choose a real orthonormal basis of the traceless Hermitian matrices. For each basis coordinate write the corresponding entries of A_i as a_i, and set $z_i = a_i/t_i$. The constraint is $t \cdot z = 0$ and the quadratic form, up to its positive scalar factor, is

$$\|z\|_2^2 - (\mathbf{1}\cdot z)^2.$$

Let $v = \mathbf{1} - nt/T$, the orthogonal projection of $\mathbf{1}$ onto $t^\perp$. On that subspace the representing operator is $I - vv^\mathsf{T}$. Since $\|v\|_2^2 = n - n^2/T$, its eigenvalues are 1 with multiplicity $n-2$ and $\lambda(t)$ once. When $t = \mathbf{1}$, the operator is the identity and the same eigenvalue list remains valid. Each basis coordinate gives one copy of this form. The change from a to z is invertible, so Sylvester's law of inertia proves the result. These displayed eigenvalues refer to the z-coordinate Euclidean metric, not to the unweighted tuple Frobenius metric. □

Proof of Theorem 6.1. Since $T = n + \sum_i (t_i - 1)^2$, the strict inequality in (6.1) is equivalent to $\lambda(t) > 0$. In this case Lemma 6.2 gives

$$\mathcal{H}_t(A) \geq m(t)\|A\|^2, \qquad m(t) = \frac{(n-2)!}{n^{n-2}}\prod_i t_i \,\frac{\min\{1,\lambda(t)\}}{(\max_i t_i)^2} > 0.$$

Taylor's theorem and stationarity yield the asserted local inequality with $a(t) = m(t)/4$, after shrinking the neighborhood. If $\lambda(t) < 0$, take $A_i = t_i v_i H$, where $v = \mathbf{1} - nt/T$ and H is any nonzero traceless Hermitian matrix. Then $A \in \mathcal{E}$ and (6.2) is strictly negative. Relative interior feasibility excludes local minimality.

It remains to treat the degenerate case $T = n^2/(n-1)$. Define

$$z_i = 1 - \frac{n-1}{n}t_i.$$

Direct calculation gives

$$\sum_i z_i = \sum_i z_i^2 = 1, \qquad \sum_i t_i z_i = 0, \qquad z_i < 1 \quad \text{for every } i.$$

Writing $e_k(z)$ for the elementary symmetric polynomials, these relations imply $e_2(z) = 0$ and

$$e_3(z) = \frac{\sum_i z_i^3 - 1}{3} < 0. \tag{6.3}$$

Indeed $z_i^3 \leq z_i^2$ with strict inequality whenever $z_i \neq 0$, and at least one z_i is nonzero. Newton's identities give the equality in (6.3).

For $n \geq 3$ take $H = \operatorname{diag}(n-1, -1, \ldots, -1)$ and $A_i = t_i z_i H$. This commuting direction belongs to $\mathcal{E}$. Its first two variations vanish. The coefficient of s^3 in $D(X^0(t) + sA)$ is

$$\frac{2(n-3)!}{n^{n-3}}\prod_i t_i \, e_3(z)\operatorname{tr}(H^3) < 0. \tag{6.4}$$

To verify the coefficient, multilinearity groups the terms indexed by three-element subsets, while polarization of the cubic determinant identity gives $D(H,H,H,I,\ldots,I) = 2(n-3)!\operatorname{tr}(H^3)$ for traceless H. Finally $\operatorname{tr}(H^3) = n(n-1)(n-2) > 0$. The remaining terms have degree at least four. Thus all sufficiently small positive s produce feasible positive definite tuples with a strictly smaller mixed discriminant. For $n = 2$, positivity and $t_1 + t_2 = 2$ imply $\sum_i (t_i - 1)^2 < 2$, so the degenerate case cannot arise. □

Theorem 6.3 (Uniform global growth near balanced traces)**.** *For each fixed integer $n \geq 2$, there are constants $\epsilon_n > 0$ and $\kappa_n > 0$ such that, for every positive t with $\sum_i t_i = n$ and $\|t - \mathbf{1}\|_1 < \epsilon_n$, and every $X \in \mathcal{F}_t$,*

$$D(X) - \frac{n!}{n^n} \prod_i t_i \geq \kappa_n \sum_i \left\| X_i - \frac{t_i}{n} I_n \right\|_F^2 . \tag{6.5}$$

For $n \geq 3$, one may take $\epsilon_n = 10^{-9} n^{-8}$ and $\kappa_n = (n!/n^n)/(4 \cdot 10^9 n^8)$. Consequently $X^0(t)$ is the unique global minimizer on $\mathcal{F}_t$.

Proof. Let $p_X(x) = \det(\sum_i x_i X_i)$. It is normalized and real stable: if every $\operatorname{Im} x_i > 0$, the imaginary part of the pencil is positive definite because $\sum_i X_i = I_n$. Its coefficients are nonnegative and its gradient is t. For $B(x) = \sum_i x_i (X_i - t_i I_n/n)$, the centered roots are the eigenvalues of $B(x)$. Consequently

$$x^{\mathsf{T}} Q_{p_X} x = \frac{1}{2} \operatorname{tr}(B(x)^2), \qquad \operatorname{tr} Q_{p_X} = \frac{1}{2} \sum_i \|X_i - t_i I_n/n\|_F^2 .$$

Theorem 2.1 now gives the result after dividing its growth constant by two. □

Theorem 6.3 answers the fixed-dimension existence question in [5, Question 1.7]. Their mixed discriminant is $D/n!$, so the minimizers are unchanged. Their Theorem 1.2 fixes each individual trace to one and perturbs the matrix sum; here the matrix sum remains the identity and the individual traces vary. Theorem 3.1 supplies the explicit radius above; a dimension-independent radius is not established. The exact local threshold in Theorem 6.1 likewise does not assert global minimality throughout that larger region.

Corollary 6.4 (Permanent specialization)**.** *Let $\mathcal{T}_t$ be the set of nonnegative $n \times n$ matrices with row sums t and column sums $\mathbf{1}$. The rank-one matrix $C_t = (t_i/n)_{i,j}$ is a relative local minimum of the permanent on $\mathcal{T}_t$ if and only if (6.1) holds. The near-balanced global inequality (6.5) also holds with $D(X)$ replaced by* per B *and the squared tuple norm replaced by $\|B - C_t\|_F^2$.*

Proof. Apply the preceding results to the diagonal matrices $X_i = \operatorname{diag}(b_{i1}, \ldots, b_{in})$. Conversely, all the negative directions used above can be chosen diagonal, so failure of the local criterion is already witnessed within $\mathcal{T}_t$. □

7. Commuting minima and the ternary reduction

In three dimensions the matrix problem admits a more precise geometric description. The first step is a commuting-minimum theorem: every relative local minimum is simultaneously diagonalizable. It turns the global problem into the minimization of a permanent on a transportation polytope, where boundary faces can be analyzed explicitly.

For this section let $t_1, t_2, t_3 > 0$ with $t_1 + t_2 + t_3 = 3$. Write $\mathcal{M}(t)$ for the triples of complex Hermitian positive semidefinite 3×3 matrices satisfying $\sum_i X_i = I$ and $\operatorname{tr} X_i = t_i$, and put

$$\mathcal{T}(t) = \{B \in \mathbb{R}_{\geq 0}^{3\times 3} : B\mathbf{1} = t, B^{\mathsf{T}}\mathbf{1} = \mathbf{1}\}.$$

We use the full coefficient normalization $D(X_1, X_2, X_3) = [z_1 z_2 z_3] \det(\sum_i z_i X_i)$. In sums indexed by i, the indices j, k denote its two complements in $\{1, 2, 3\}$. The rank-one matrix $B^\circ_{ij} = t_i/3$ has permanent $P_0(t) = 2t_1 t_2 t_3/9$.

Lemma 7.1 (Multiplier certificate)**.** *For positive traces, at every relative local minimum there exist a Hermitian H, real μ_i, and $S_i \geq 0$ such that*

$$2X_i^2 + (1 - 2t_i)X_i = H + \mu_i I + S_i, \qquad S_i X_i = X_i S_i = 0.$$

Proof. Inclusion–exclusion and the cubic determinant formula give

$$D = C(t) + \sum_i \left\{ \frac{2}{3} \operatorname{tr} X_i^3 + \left(\frac{1}{2} - t_i \right) \operatorname{tr} X_i^2 \right\}, \quad C(t) = 1 - \frac{1}{2} \sum_i t_i^2 + \frac{1}{3} \sum_i t_i^3.$$

Write its gradient as $G_i = 2X_i^2 + (1 - 2t_i)X_i$. Convex feasible segments imply that X minimizes $\langle G, Y \rangle$ over the feasible set.

For completeness let $\mathcal{A}$ encode the linear constraints, $b = \mathcal{A}X$, and $v = \langle G, X \rangle$. The convex open set

$$\{(\mathcal{A}Y - b, \langle G, Y \rangle - v + s) : Y_i \succ 0,\ s > 0\}$$

in range $\mathcal{A} \times \mathbb{R}$ does not contain zero. Separation gives a nonzero (λ, η) with

$$\lambda \cdot (\mathcal{A}Y - b) + \eta(\langle G, Y \rangle - v + s) \geq 0.$$

Thus $\eta \geq 0$. If $\eta = 0$, variations around the strictly positive feasible tuple $Y_i = t_i I/3$ imply $\mathcal{A}^* \lambda = 0$, contradicting nonzero separation on the range. Normalize $\eta = 1$. Unbounded positive directions imply $S = G + \mathcal{A}^* \lambda \succeq 0$. Letting $Y, s \to 0$ gives $v + \lambda \cdot b \leq 0$; evaluation at $X \succeq 0$ gives the reverse inequality. Hence $\operatorname{tr} S_i X_i = 0$ for every i, which for positive semidefinite matrices implies $S_i X_i = X_i S_i = 0$. The adjoint constraint map has a common Hermitian component and individual scalar components, proving the assertion. □

Proposition 7.2. *For positive traces of sum three, every relative local minimum of D consists of pairwise commuting matrices. Consequently*

$$\min_{\mathcal{M}(t)} D = \min_{\mathcal{T}(t)} \operatorname{per}.$$

Proof. Lemma 7.1 implies $[H, X_i] = 0$. A simple-spectrum H therefore diagonalizes all X_i.

Scalar multiplier. Suppose H is scalar. Positive eigenvalues of each X_i solve one quadratic. A rank-two matrix cannot have two distinct positive eigenvalues: their sum would be $t_i - 1/2$, whereas its trace is t_i. Thus every X_i can be written $a_i I + b_i P_i$ with P_i rank one; the full-rank case follows because the quadratic has at most two roots. The sum constraint becomes $\sum_i b_i P_i = cI$. If $c \neq 0$, write $P_i = v_i v_i^*$ and $V = (v_1, v_2, v_3)$. Then V is invertible and $V \operatorname{diag}(b_i) V^* = cI$. Congruence gives $b_i/c > 0$, and $V \operatorname{diag} \sqrt{b_i/c}$ is unitary. Thus the projections are mutually orthogonal. If $c = 0$, split positive and negative coefficients. If all coefficients vanish the tuple is scalar. Otherwise both signs are present, and one side has only one term; equality of positive semidefinite sums forces every participating projection to have the same range. All X_i commute.

A double eigenvalue. It remains to treat multiplicities $1 + 2$ of H. Write $X_i = a_i \oplus B_i$, $\sum_i a_i = 1$, and $\sum_i B_i = I_2$. Set $b_i = \operatorname{tr} B_i$ and $B_i = b_i I_2/2 + V_i$, with $\sum_i V_i = 0$ and $\|V_i\|^2 = \operatorname{tr} V_i^2/2$. Positivity means $\|V_i\| \leq b_i/2$. Block determinant expansion gives

$$D = \frac{1}{2} \sum_i a_i b_j b_k + \sum_i (1 - 2a_i) \|V_i\|^2.$$

If all $a_i \leq 1/2$, contraction toward zero shows that every V_i with positive coefficient must vanish. There can be at most two zero coefficients since $\sum_i a_i = 1$. Any remaining two vectors are opposite, so commute. Otherwise there is exactly one $a_k > 1/2$. If $V_k = 0$, the other two are opposite. If $V_k \neq 0$, write $V_i = U_i + W_i$, where U_i lies in the real line spanned by V_k and W_i is perpendicular to it for the inner product $\operatorname{tr}(UV)/2$. Replace V_i by $U_i + sW_i$ for $s < 1$ sufficiently close to one. The sum and positivity constraints are preserved, V_k is fixed, and every nonzero perpendicular component strictly decreases D. Thus all V_i are collinear at a local minimum and commute.

Compactness gives a global minimum. Simultaneous diagonalization identifies D with a permanent, and diagonal tuples realize every matrix in $\mathcal{T}(t)$. □

8. The sharp Euclidean trace ball

The polytope reduction makes the first global competitors visible. The Euclidean ball is determined by a one-zero face and its boundary; the proof excludes every other zero pattern or gives its exact gap.

Theorem 8.1. *Let $t_1 + t_2 + t_3 = 3$ and $\rho^2 = \sum_i (t_i - 1)^2 \le 3/8$. For every $B \in \mathcal{T}(t)$,*

$$\operatorname{per} B \ge P_0(t).$$

If $\rho^2 < 3/8$, equality holds only at B°. On the closed ball, every other equality case is, up to row and column permutations,

$$t = (3/2, 3/4, 3/4), \qquad B_c = \begin{pmatrix} 0 & 3/4 & 3/4 \\ 1/2 & c & 1/4 - c \\ 1/2 & 1/4 - c & c \end{pmatrix}, \qquad 0 \le c \le 1/4.$$

The Euclidean radius $\sqrt{3/8}$ is sharp: every larger centered ball contains a trace vector admitting a strict violation.

Proof. The trace sum and Cauchy–Schwarz give $\rho^2 \ge 3(t_i - 1)^2/2$, hence

$$1/2 \le t_i \le 3/2.$$

If one trace equals $3/2$, the other two equal $3/4$. Choose a global minimizer B. It lies in the relative interior of its smallest face, so the objective is stationary along that face and its face Hessian is positive semidefinite. We classify the possible zero patterns. Row and column permutations are used to put each face in a convenient chart.

No zeros Write $B = B^\circ + A$. All row and column sums of A vanish, and

$$\operatorname{per}(B^\circ + sA) = P_0 + qs^2 + cs^3, \qquad q = \frac{1}{6} \sum_i (3 - 2t_i) \|A_{i\cdot}\|^2.$$

Indeed the quadratic coefficient is $-\sum_i t_i \langle A_{j\cdot}, A_{k\cdot} \rangle / 3$. For nonzero A, $q > 0$: at most one coefficient vanishes, and a single nonzero row is incompatible with zero column sums. If B is positive and nonscalar, stationarity at $s = 1$ gives $2q + 3c = 0$, but the second derivative is $-2q < 0$. Thus only B° can minimize in the interior.

Two zeros in one row Take the row to be $(0, 0, t_1)$; then $t_1 \le 1$. The remaining 2×2 block has column sums $1, 1$ and a row sum $r \in [2 - t_3, t_2]$. Its permanent is

$$2(x - r/2)^2 + r(2 - r)/2.$$

Concavity in r gives

$$\operatorname{per} B \ge \frac{t_1}{2} \min\{t_2(2 - t_2), t_3(2 - t_3)\}.$$

Each endpoint minus P_0 equals

$$\frac{t_1 t_j}{18}(18 - 9t_j - 4t_k).$$

Since $9t_j + 4t_k = 5t_j + 4(3 - t_1) \le 35/2$, this difference is at least $1/144$.

Two zeros in one column The column is $(1,0,0)^\top$. Write

$$B = \begin{pmatrix} 1 & a & t_1 - 1 - a \\ 0 & b & t_2 - b \\ 0 & 1 - a - b & t_3 - 1 + a + b \end{pmatrix}.$$

Here $1 \le t_1 \le 3/2$ and $0 \le a \le t_1 - 1$, and

$$\operatorname{per} B = t_2(1-a) + (t_3 - t_2 - 2 + 2a)b + 2b^2.$$

Unrestricted minimization over b gives the lower bound

$$F(a) = t_2(1-a) - (t_3 - t_2 - 2 + 2a)^2/8.$$

This is concave in a with equal endpoint values, so

$$\operatorname{per} B \ge (2 - t_1)/2 - (t_2 - t_3)^2/8.$$

Put $u = 2 - t_1 \in [1/2, 1]$. Then $|t_2 - t_3| \le u$ and $P_0 \le (2-u)(1+u)^2/18$. Consequently

$$\operatorname{per} B - P_0 \ge (4u^3 - 9u^2 + 24u - 8)/72 \ge 1/32.$$

The cubic is increasing on this interval and equals $9/4$ at $u = 1/2$. This argument includes all edges of the face.

Matching zeros For three diagonal zeros,

$$B(x) = \begin{pmatrix} 0 & x & t_1 - x \\ 2 - t_3 - x & 0 & 1 - t_1 + x \\ t_3 - 1 + x & 1 - x & 0 \end{pmatrix}.$$

Direct expansion gives

$$\operatorname{per} B(x) - P_0 = 3(x - x_*)^2 + \frac{1}{36}\prod_i (3 - 2t_i), \quad x_* = \frac{1}{2} + \frac{t_1 - t_3}{3}.$$

This is nonnegative. Equality forces the stated exceptional trace profile and yields endpoints of B_c.

Exactly one zero Parameterize

$$B = \begin{pmatrix} 0 & a & t_1 - a \\ b & c & t_2 - b - c \\ 1 - b & 1 - a - c & t_3 - 2 + a + b + c \end{pmatrix}.$$

Then

$$P = at_2 + t_1 b + (2t_3 - 7)ab + 2a^2 b + 2ab^2 + c(2a - t_1)(2b - 1).$$

All other entries are positive. A positive semidefinite face Hessian with $P_{cc} = 0$ must satisfy $P_{ac} = P_{bc} = 0$. Thus $a = t_1/2$, $b = 1/2$. Stationarity in b gives $t_2 = t_3$, and

$$P - P_0 = t_1^2(3 - 2t_1)/36 \ge 0.$$

Equality gives B_c with $0 < c < 1/4$.

Exactly two separated zeros Write

$$B = \begin{pmatrix} 0 & a & t_1 - a \\ b & 0 & t_2 - b \\ 1-b & 1-a & d \end{pmatrix}, \quad d = t_3 - 2 + a + b.$$

All other entries are positive, so $a, b > 0$ and $0 < d < 1$. Let $K = 3 - 2d$ and $z = 2 - 2(a + b)$. Face stationarity gives

$$t_1 = a(K - 2b), \quad t_2 = b(K - 2a), \quad 4ab = (1 - d)(1 - z), \quad t_3 = 1 + d + z/2.$$

It follows that $\rho^2 = R(d, z)$, where

$$2R = (3 - 3d + d^2)z^2 + (-6d + 12d^2 - 4d^3)z + 9d - 9d^2 + 4d^3,$$

and

$$P - P_0 = \frac{ab}{9}H, \quad H = 1 - 8d + 10d^2 - 8d^3 + (1 + 7d - 2d^2)z - (2 - d)z^2.$$

First, $2R - 3d$ is a quadratic in z with leading coefficient $A_0 = 3 - 3d + d^2 > 0$, linear coefficient $L = -6d + 12d^2 - 4d^3$, and constant coefficient $C_0 = 6d - 9d^2 + 4d^3$. Its discriminant is

$$L^2 - 4A_0C_0 = 4d(d - 1)Q(d), \quad Q(d) = 18 - 36d + 45d^2 - 24d^3 + 4d^4.$$

For $u = 1 - d \in [0, 1]$, $Q(1 - u) = 7 + 2u - 3u^2 + 8u^3 + 4u^4 \geq 4$. Thus $R > 3d/2$, and $R \leq 3/8$ implies $d < 1/4$.

Put $g = 3/8 - R \geq 0$. The exact identity

$$H = (2 + 2d)g + \alpha z^2 + \beta z + \gamma$$

holds with

$$\alpha = 1 + d - 2d^2 + d^3, \quad \beta = 1 + d + 4d^2 + 8d^3 - 4d^4, \quad 4\gamma = 1 + d + 40d^2 - 52d^3 + 16d^4.$$

For $0 < d \leq 1/4$, $\alpha \geq 7/8$ and

$$\beta^2 - 4\alpha\gamma = d^2\Psi(d),$$

$$\Psi(d) = -30 + 37d + 139d^2 - 104d^3 + 116d^4 - 80d^5 + 16d^6 \leq -\frac{2971}{256} < 0.$$

The last inequality drops negative terms and substitutes $d = 1/4$ in positive terms. Hence the quadratic is strictly positive for every real z, so $H > 0$ and $P > P_0$.

Exhaustion and equality With at least three zeros, either a row or column contains two, or exactly three zeros form a perfect matching. Together with the zero-, one-, and two-zero cases, this exhausts all global minima. The strict cases exclude every additional equality pattern. The matching and one-zero identities give exactly B_c and the exceptional trace profile. These calculations prove the inequality and its complete equality classification.

Sharpness. For $3/2 < x < 2$, let

$$B_x = \begin{pmatrix} 0 & x/2 & x/2 \\ 1/2 & (2-x)/4 & (2-x)/4 \\ 1/2 & (2-x)/4 & (2-x)/4 \end{pmatrix}.$$

Its row sums are $(x, (3 - x)/2, (3 - x)/2)$, and

$$\rho^2 = \frac{3}{2}(x - 1)^2, \qquad \operatorname{per} B_x - P_0 = \frac{x^2(3 - 2x)}{36} < 0.$$

Letting $x \downarrow 3/2$ proves sharpness.

□

9. The sharp ℓ^1 trace ball

For the ℓ^1 metric the first competitor lies on a different face. The preceding exclusions reduce the problem to two separated zeros. On that face, a stationary chart expresses both the coefficient gap and the marginal distance in two variables. The sharp radius is then the minimum distance on the zero-gap curve. Set

$$\varepsilon_0 = \sqrt{\frac{236 - 37\sqrt{37}}{12}}, \quad d_* = \frac{\sqrt{37} - 5}{12}, \quad s_* = \sqrt{d_* + 3d_*^2}, \quad m_* = \varepsilon_0/2.$$

For $s = \sqrt{d + 3d^2}$ write

$$B_d = \begin{pmatrix} 0 & (1+s+d)/2 & (1-d)(1+2s)/2 \\ (1-s+d)/2 & 0 & (1-d)(1-2s)/2 \\ (1+s-d)/2 & (1-s-d)/2 & d \end{pmatrix}.$$

Theorem 9.1. *If $\sum_i t_i = 3$ and $\|t - \mathbf{1}\|_1 \le \varepsilon_0$, then every $B \in \mathcal{T}(t)$ satisfies* $\operatorname{per} B \ge P_0$. *Inside the ball equality holds only at B°. Additional closed-ball equality cases occur only at permutations of $(1 + m_*, 1 - m_*, 1)$; for that ordering they are the six column permutations of B_{d_*}. Every larger radius admits strict violations.*

The face calculations in the preceding proof apply throughout $1/2 \le t_i \le 3/2$, apart from the special estimate on the two-zero face. Since zero-sum triples satisfy $\|t - \mathbf{1}\|_1 = 2\max_i |t_i - 1|$ and $0 < \varepsilon_0 < 1$, the present ball lies in the strict box $1/2 < t_i < 3/2$. There the matching-zero and one-zero gaps are strictly positive. Consequently every nonscalar global minimizer with value at most P_0 has exactly two zeros in distinct rows and columns. We analyze that face before completing the proof of Theorem 9.1.

9.1. A stationary chart for the two-zero face

Write

$$B = \begin{pmatrix} 0 & a & t_1 - a \\ b & 0 & t_2 - b \\ 1 - b & 1 - a & d \end{pmatrix}, \quad d = t_3 - 2 + a + b.$$

All seven other entries are positive. With $K = 3 - 2d$, stationarity gives

$$t_1 = a(K - 2b), \quad t_2 = b(K - 2a), \quad 4ab = (1 - d)(2(a + b) - 1).$$

Since $0 < d < 1$, the last equation implies $a + b > 1/2$. Setting $A = 1 - 2a$, $C = 1 - 2b$ therefore gives $AC = d(A + C - 1) < 0$. After swapping the zero rows and columns if necessary,

$$a = (1 + v)/2, \quad b = (1 - u)/2, \quad d = \frac{uv}{1 + v - u}, \quad u, v > 0, \quad u + v < 1.$$

Indeed positivity of $t_2 - b$ is equivalent to

$$1 - v - 2d = \frac{(1 + v)(1 - u - v)}{1 + v - u} > 0;$$

the converse also gives all required positive entries. In particular $u, v < 1$, $1 + v - u > 0$, and $u - 2d = u(1 - u - v)/(1 + v - u) > 0$. Since $u - 2d > 0$, $t_1 > 1$ and $t_2 < 1$.

Put

$$h = d + (u - v)/2, \quad S = h^2 - 4dh + 3d^2 + d, \quad w = \sqrt{S} = (u + v)/2.$$

Expansion gives

$$P - P_0 = \frac{ab}{9} H(d,h), \quad H = 1 - 10d - 12d^2 + (2 + 30d - 12d^2)h - 4(2-d)h^2,$$

and the exact trace distance is $E = (3-2d)\sqrt{S} + |h|$. Thus H records the sign of the objective gap, while E records the distance from the balanced marginal. The task is to minimize E over the part of the chart where $H \le 0$.

9.2. Sharp distance inequality

Lemma 9.2. *On this chart, $H \le 0$ implies $E \ge \varepsilon_0$. Equality requires $d = d_*$ and $h = 0$.*

Proof. Restricting the parameter range. Assume $E < 1$, since otherwise the claim is immediate. The chart gives $0 < d < 1/2$ and the trace distance gives $|h| < 1/2$. With $S = (h-2d)^2 + d(1-d)$ and $K = 3 - 2d$, Cauchy–Schwarz gives

$$E \ge K\sqrt{S} + h \ge 2d + \sqrt{(K^2-1)d(1-d)}.$$

For $1/3 \le d < 1/2$, we have $K^2 - 1 \ge 3$ and $d(1-d) \ge 2/9$, so the right side is at least $2/3 + \sqrt{2/3} > 1$. Hence $0 < d < 1/3$.

Reduction to the zero-gap curve. Let $r(d)$ be the lower root of $H(d,h) = 0$ and $\ell(d) = (9d - 1)/4$. The identities

$$H(d,-1/4) = -69d/4 - 9d^2, \quad H(d,1/2) = 6d(1-3d),$$

$$H(d,\ell) = -3d(3d-1)(3d-5)/4, \quad H(d,3d/2) = (1-d)(1-3d)^2$$

give $-1/4 \le \ell \le r \le 3d/2 \le 1/2$, $r(0) = -1/4$, and $r(1/3) = 1/2$. At the lower root $H_h > 0$. Since H_d increases in h on the relevant rectangle,

$$H_d(d,r) \le H_d(d,3d/2) = -10 + 21d - 27d^2 \le -6.$$

Thus $r' = -H_d/H_h > 0$.

Under $|h| < 1/2$, $H \le 0$ means $h \le r(d)$. At fixed h,

$$E_d = \frac{T}{2\sqrt{S}}, \quad T = 3 + 12d - 24d^2 - 12h + 24dh - 4h^2.$$

Here T decreases in h for $-1/2 \le h \le 3d/2$, and $T(d,3d/2) = 3(1-d)^2 > 0$. If $h \le -1/4$, decreasing d to zero gives $E(d,h) > 4|h| \ge 1$, a contradiction. Otherwise there is a unique $D \le d$ with $r(D) = h$, and $E(d,h) \ge E(D,r(D))$, strictly unless $D = d$.

Minimization along the curve. It remains to determine where the distance changes from decreasing to increasing. Define

$$N = K(S_d H_h - S_h H_d) - 4SH_h, \qquad W = 4SH_d^2 - N^2.$$

For $J(d) = K\sqrt{S(d,r(d))}$,

$$2\sqrt{S}\,H_h J' = N, \qquad (J + |r|)' = \frac{N - 2\,\mathrm{sgn}(r)\sqrt{S}\,H_d}{2\sqrt{S}\,H_h}.$$

On $H = 0$ the exact identity

$$W = \frac{4d}{(2-d)^3}\big(A(d) + 2hB(d)\big)$$

holds, with coefficients given in Section A. This identity separates the algebraic positivity check from the geometric argument: positivity of W fixes the sign of the distance derivative on either side of $r = 0$. For $\ell \le h \le 1/2$, the affine expression is positive because its endpoint polynomials $F_0 = A + (9d-1)B/2$ and $F_1 = A + B$ have the strictly positive Bernstein coefficients in Section A. Thus $W > 0$ and $|N| < 2\sqrt{S}(-H_d)$. The derivative of $J + |r|$ is strictly negative for $r < 0$ and strictly positive for $r > 0$. Its unique minimum occurs at $r = 0$, where $1 - 10d - 12d^2 = 0$, so $d = d_*$. Finally,

$$E(d_*, 0)^2 = (3 - 2d_*)^2(d_* + 3d_*^2) = (236 - 37\sqrt{37})/12.$$

All strictness statements give the asserted equality condition. □

9.3. Completion and sharpness

A nonscalar global minimum in the closed ball must lie on the stationary face and have $H \le 0$. Lemma 9.2 forces equality in distance, $h = 0$, and $d = d_*$. The stationary equations then recover B_{d_*} and traces $(1 + m_*, 1 - m_*, 1)$. Its deficit is zero, proving Theorem 9.1.

For B_d with $s = \sqrt{d + 3d^2}$,

$$t = (1 + m(d), 1 - m(d), 1), \quad m(d) = (3 - 2d)s/2,$$

$$\operatorname{per} B_d - P_0 = \frac{(1-d)(1+2d)(1-10d-12d^2)}{36}.$$

Taking $d > d_*$ tending to d_* gives strict violations with distance tending to ε_0. A rational version sets $d = 1/(\eta^2 - 3)$, $s = \eta/(\eta^2 - 3)$. The deficit has the sign of $\eta^4 - 16\eta^2 + 27$. Rational $\eta > 3$ approaching $\sqrt{8 + \sqrt{37}}$ from below give fully rational sharpness examples.

10. Mixed-discriminant and stable-cubic consequences

The permanent results now return to the matrix and polynomial settings. Commuting minima transfer both the bounds and their equality cases to Hermitian triples. A definite ternary representation then gives the full stable-cubic statement.

Corollary 10.1. *Both Theorems 8.1 and 9.1 hold with* per B *replaced by* $D(X_1, X_2, X_3)$ *on* $\mathcal{M}(t)$. *Inside either respective open trace ball, equality holds only at* $X_i = t_i I/3$. *The additional boundary minimizers are precisely the simultaneous unitary conjugates of diagonal tuples whose diagonal-entry rows form the corresponding matrices* B_c *or* B_{d_*}, *allowing row relabeling. For the normalized mixed discriminant* $D/6$, *all values in these statements are divided by six.*

Proof. Proposition 7.2 identifies the minimum with the permanent minimum. Every equality tuple is a global, hence local, minimum and therefore commutes. Simultaneous diagonalization and the complete permanent equality classifications give the assertion. Diagonal tuples also preserve both sharpness constructions. □

Corollary 10.2. *Let* f *be a homogeneous ternary real-stable polynomial of degree three with* $f(\mathbf{1}) = 1$ *and* $\nabla f(\mathbf{1}) = t$. *Under either trace-ball condition,*

$$[z_1 z_2 z_3] f \ge 2t_1 t_2 t_3/9.$$

Inside either open ball the unique minimizing polynomial is $((t_1 z_1 + t_2 z_2 + t_3 z_3)/3)^3$. *Additional boundary minimizing polynomials are exactly the products of column linear forms of the matrices* B_c *in the Euclidean case, or of* B_{d_*} *in the* ℓ^1 *case, with the same trace relabelings. Both radii are sharp for this class.*

Proof. A real polynomial is real stable when it has no zero with every variable in the open upper half-plane. Homogeneity shows that $f(v) \neq 0$ for every $v > 0$: otherwise $f(iv) = i^3 f(v) = 0$. It also shows hyperbolicity in the direction $\mathbf{1}$: a nonreal root of $s \mapsto f(x + s\mathbf{1})$ would, after complex conjugation if necessary, contradict stability. The definite determinantal representation theorem for ternary hyperbolic forms [6] gives real symmetric matrices representing f, with a positive definite pencil value at $\mathbf{1}$. Congruence normalization, using $f(\mathbf{1}) = 1$, gives

$$f(z) = \det\Big(\sum_{i=1}^{3} z_i X_i\Big), \qquad \sum_i X_i = I.$$

Along the connected positive orthant the pencil remains nonsingular, so its inertia is constant and positive. Taking limits as $v > 0$ approaches the ith coordinate vector gives $X_i \geq 0$. Differentiation at $\mathbf{1}$ gives $\operatorname{tr} X_i = t_i$. Corollary 10.1 proves the bound. If equality holds, this representing tuple is a matrix global minimum, so it commutes; the permanent equality classifications then give exactly the stated products. Conversely each displayed product has nonnegative linear factors, is stable, and has the required normalization and gradient. The diagonal sharpness examples belong to this subclass. □

11. Escape height and its attainment

The preceding results identify when the scalar point is locally or globally minimizing. A path question asks what height is needed to leave a prescribed quadratic sublevel set. On the symmetric marginal family below, the crossing height can be computed exactly. Its value is continuous along the family, but its attainment changes at the balanced marginal.

The path space here is a transportation polytope of nonnegative matrices. This distinction matters: the theorem does not concern arbitrary noncommuting Hermitian paths, and its target is an open set rather than a prescribed endpoint.

Let $a, b > 0$ satisfy $a + 2b = 3$ and $a \geq b/4$, and put

$$c = (a, b, b), \qquad t = \big(b, (a+b)/2, (a+b)/2\big), \qquad C_{ij} = t_i/3.$$

Write $\mathcal{T}(t) = \{X \geq 0 : X\mathbf{1} = t, X^{\top}\mathbf{1} = \mathbf{1}\}$ and $\mathcal{V} = \{A : A\mathbf{1} = A^{\top}\mathbf{1} = 0\}$. For $A \in \mathcal{V}$, define

$$q(A) = \frac{1}{6}\sum_{i=1}^{3} c_i \|A_{i\bullet}\|_2^2, \qquad F(A) = \operatorname{per}(C + A) - \operatorname{per} C.$$

Set $Q = ab^2/12$, $\Delta = ab^2/36$, and $\mathcal{E} = \{X \in \mathcal{T}(t) : q(X - C) > Q\}$. The escape height is

$$H = \inf_{\substack{\gamma \in C([0,1], \mathcal{T}(t)) \\ \gamma(0) = C,\ \gamma(1) \in \mathcal{E}}} \max_{0 \leq u \leq 1} \big(\operatorname{per} \gamma(u) - \operatorname{per} C\big).$$

The target set is open relative to the polytope, so attainment is a separate issue from the value of this infimum.

Theorem 11.1 (Symmetric-marginal escape and nonattainment)**.** *Under the preceding assumptions, $\mathcal{E} \neq \varnothing$ and $H = \Delta$. If $a \neq b$, an explicit path attains this height. If $a = b = 1$, no path attains it. Thus the infimum is continuous along this family, whereas attainment fails at its balanced member.*

Proof. The lower bound. The quadratic part of the permanent and a weighted arithmetic–geometric mean inequality determine a crossing level that every escape path must meet. Expanding the permanent at C gives

$$F(A) = q(A) + \operatorname{per} A \qquad (A \in \mathcal{V}). \tag{11.1}$$

Indeed, the linear term vanishes by the zero row sums; the quadratic term is $-\frac{1}{3}\sum_{i<j} t_k\langle A_i, A_j\rangle$, where $\{i, j, k\} = \{1, 2, 3\}$. Since $\sum_i A_i = 0$ and $c_i + c_j = 2t_k$, this is $q(A)$.

Use the orthonormal basis $e = (2, -1, -1)/\sqrt{6}$, $f = (0, 1, -1)/\sqrt{2}$ of the zero-sum row plane. Write $A_i = \Re(z_i)e + \Im(z_i)f$, so that $\sum_i z_i = 0$. Direct expansion gives

$$\operatorname{per} A = \sqrt{2/3}\,\Re(z_1z_2z_3), \qquad 6q(A) = \sum_i c_i|z_i|^2.$$

Weighted arithmetic–geometric mean therefore yields

$$|\operatorname{per} A| \le \frac{4}{\sqrt{3ab^2}}q(A)^{3/2}. \tag{11.2}$$

Every admissible escape path crosses $q = Q$, where (11.1)–(11.2) give $F \ge Q/3 = \Delta$. Hence $H \ge \Delta$.

An attaining path away from balance. For the upper bound, set

$$d = \frac{2a+b}{12}, \qquad v = \sqrt{\frac{b(4a-b)}{48}}, \qquad S = \begin{pmatrix} 0 & b/2 & b/2 \\ 1/2 & d+v & d-v \\ 1/2 & d-v & d+v \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Here $v \ge 0$, and

$$d^2 - v^2 = \frac{(a-b)^2}{36}. \tag{11.3}$$

Consequently $S \in \mathcal{T}(t)$, and its last two rows are strictly positive when $a \ne b$. With $A = S - C$, direct substitution in the six permanent monomials and in q gives

$$q(A) = Q, \qquad \operatorname{per} A = -2Q/3, \qquad F(A) = \Delta, \tag{11.4}$$

$$\operatorname{per}(S + zK) = bd = \operatorname{per} S, \qquad q(A + zK) = Q + \frac{4bv}{3}z + \frac{2b}{3}z^2. \tag{11.5}$$

In particular, $F(\lambda A) = \Delta(3\lambda^2 - 2\lambda^3)$. If $a \ne b$, choose $0 < z < d - v$. The radial segment from C to S, followed by the segment from S to $S + zK$, is feasible. Its maximum excess permanent is exactly Δ, and its endpoint has $q > Q$. This includes $a = b/4$, where $v = 0$ and the positive quadratic term in (11.5) supplies the strict escape.

The balanced case. It remains to analyze $a = b = 1$. Then $t = \mathbf{1}$, $Q = 1/12$, $\Delta = 1/36$, and a column permutation of S is

$$S_0 = (J - I)/2.$$

Let

$$L = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

For $|z| \le 1/2$, the matrix $S_0 + zL$ is doubly stochastic, and

$$q(S_0 - C + zL) = 1/12 + z^2, \qquad \operatorname{per}(S_0 + zL) = 1/4 + 3z^2. \tag{11.6}$$

The radial segment $C \to S_0$, followed by a segment with any $0 < |z| \le 1/2$, is an escape path of height $\Delta + 3z^2$. Letting $z \to 0$ proves $H = \Delta$ and $\mathcal{E} \ne \varnothing$.

To prove nonattainment, we identify all possible crossing points at the minimal height and show that none can be followed by a feasible escape at that height. Equality in (11.2) at $q = 1/12$, with negative permanent, requires $|z_1| = |z_2| = |z_3|$ and $z_1z_2z_3 < 0$. The zero-sum complex vectors are an equilateral triangle. Its two orientations and three allowed common phases give exactly six

matrices, namely the column permutations of $S_0 - C$. Thus every crossing of $q = 1/12$ at height at most Δ must be at a column permutation of S_0.

There is a neighborhood of S_0 in which no feasible point with $q > 1/12$ has permanent at most $1/4$. To see this rigorously, suppose contrariwise that $S_0 + H_m$ is such a sequence, with $H_m \neq 0$ and $H_m \to 0$. Put $h_{mi} = (H_m)_{ii}$. Feasibility gives $h_{mi} \geq 0$. The zero marginal sums give

$$q(S_0 - C + H_m) - 1/12 = \frac{\|H_m\|_F^2 - \sum_i h_{mi}}{6}. \tag{11.7}$$

After passing to a subsequence, $H_m/\|H_m\|_F \to U \in \mathcal{V}$ with $\|U\|_F = 1$. The strict positivity of (11.7) implies $\sum_i h_{mi} < \|H_m\|_F^2$, so U has zero diagonal. Every zero-marginal 3×3 matrix with zero diagonal is a multiple of L; hence $U = \pm L/\sqrt{6}$.

The permanent is stationary at S_0 on the full affine marginal space. Indeed, every permanent cofactor of S_0 equals $1/4$, so its gradient pairs to zero with every zero-marginal displacement. Let $B(H)$ be the quadratic term of the Taylor expansion there. Equation (11.6) gives $B(zL) = 3z^2$. It follows that

$$\frac{\operatorname{per}(S_0 + H_m) - 1/4}{\|H_m\|_F^2} \longrightarrow B(U) = 1/2 > 0,$$

contradicting the assumed permanent bound. Column permutations give the same neighborhood property at each of the six gates.

Finally, a path of height at most Δ with endpoint in $\mathcal{E}$ would have a first exit time $\tau = \inf\{u : q(\gamma(u) - C) > Q\}$. Continuity gives $\tau > 0$ and $q(\gamma(\tau) - C) = Q$, hence at one of the six gates. There are later points with $q > Q$ arbitrarily close to that time. The neighborhood property contradicts the height bound. This proves nonattainment. □

Remark 11.2. The nonattainment statement depends on the open target $q > Q$. It does not claim that the scalar matrix and every prescribed endpoint have the same barrier. The flat continuation (11.5) is also a path in a commuting, nonnegative matrix model; no assertion about all Hermitian-tuple paths follows without a separate path-space argument.

12. Further questions

The first question is the dimension dependence of the global neighborhood. The proof gives an inverse-polynomial radius, while the exact local criterion places an upper constraint on the global minimization region. Determining the scale between these bounds, or finding a dimension-independent sufficient radius, would clarify how balanced rigidity governs global behavior. The coefficient-distance estimate raises a related question: can its dimension dependence be improved by a sharper comparison between the exponent law and root variance?

The cubic analysis suggests a more geometric direction. The two sharp centered balls locate the first boundary competitors in their respective metrics, but do not describe the entire set of gradients for which the scalar power minimizes globally. A description of that region would connect the stationary face charts to the local stability boundary. The escape theorem concerns an open target in a transportation polytope; paths between prescribed endpoints, or paths of noncommuting Hermitian tuples, lead to different variational problems.

Finally, the root-variance argument uses real-rooted restrictions to control all higher central coefficients. An extension to strongly log-concave polynomials would require a replacement for that control. The balanced uniqueness theorem in that larger class does not by itself supply the near-balanced quantitative estimates proved here.

A. An exact certificate for the ℓ^1 radius

Let $A(d) = \sum_{j=0}^{10} a_j d^j$ and $B(d) = \sum_{j=0}^{10} b_j d^j$, with

$$\begin{aligned}(a_j) &= (1570, -13777, 68559, -290171, 821303, -1536384,\\ &\qquad 1879620, -1471392, 704448, -186624, 20736),\\ (b_j) &= (-34, -1771, 29751, -110731, 225671, -286998,\\ &\qquad 230640, -112896, 30528, -3456, 0).\end{aligned}$$

Direct polynomial expansion gives, modulo $H = 0$,

$$(2 - d)^3 W = 4d(A + 2hB).$$

For $F_0 = A + (9d - 1)B/2$, $F_1 = A + B$, write

$$F_j(d) = \sum_{k=0}^{10} \beta_{jk} \binom{10}{k} (3d)^k (1 - 3d)^{10-k}.$$

The coefficients are:

k	β_{0k}	β_{1k}
0	1587	1536
1	69131/60	15266/15
2	44833/54	100198/135
3	26551/45	316271/540
4	3510302/8505	4133324/8505
5	3888835/13608	26000/63
6	20377997/102060	1280093/3645
7	1425187/9720	1430671/4860
8	3935542/32805	7829188/32805
9	138394/1215	221212/1215
10	10000/81	10000/81

Every entry is positive. The Bernstein basis is nonnegative and sums to one on $0 \le 3d \le 1$. Thus both endpoint polynomials are strictly positive there.

Reproducibility

The source archive contains the complete LaTeX source, the fixed style file, and seven exact-arithmetic checkers. The certificate checker verifies the polynomial identity and Bernstein coefficients printed in the appendix. The remaining scripts verify the specified matrix, root-variance, variable-replication, logarithmic-scaling, and path identities, together with the stated finite tests of dimension-dependent constants. All inputs are included. These calculations provide reproducible checks of the algebra; the general statements are established by the analytic proofs above. No formal proof-assistant verification is claimed.

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